Article

Sizing Rooftop Photovoltaics and Batteries After Energy Efficiency: An Optimisation Framework for Manufacturing SMEs

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Abstract: Small and medium-sized manufacturers are often offered rooftop photovoltaics (PV) as their first decarbonisation step, although cheaper demand-side measures could reduce the supply to be built. This study develops a modelling and optimisation framework that appraises both kinds of option on common indicators and applies it to a Mexican safety-footwear plant. A tariff model built from official charges reproduces 28 monthly bills within 2.2% and values each avoided kWh at 2.64 MXN. Four demand-side measures save 23% of grid electricity at 0.21–1.04 MXN kWh^{-1}. Sizing PV after efficiency reaches the same 54% cut in grid purchases with 94.9 instead of 163.8 kWp and 13% less capital, with a higher net present value in 97% of 5,000 Monte Carlo draws. An hourly design search verified against EnergyPlus and a linear programme in the open model urbs agree that annual net-zero electricity is profitable, whereas deep hourly self-sufficiency needs batteries that current tariffs do not repay. Under caps on grid emissions, urbs completes the efficiency bundle before adding storage, which lowers annualised costs by 29–35% versus supply-only designs. Efficiency also keeps its value if PV loses net-metering demand credit. An open toolkit generalises the framework to other SMEs.

Keywords: energy systems modelling; optimisation; energy efficiency first; industrial SMEs; rooftop photovoltaics; battery storage; Monte Carlo simulation; linear programming; EnergyPlus; Mexico

1. Introduction

Decarbonising industry while protecting firms from volatile energy markets has become a shared objective of climate and energy-security policy. The European Union has made energy efficiency first (EE1st) a legal principle: decision-makers must consider cost-effective demand-side alternatives before investing in supply, networks or markets [1–3]. Global progress nonetheless remains slow. Energy intensity improved by about 1% in 2024, far below the doubling pledged at COP28, and governments increasingly justify efficiency policy on grounds of security, affordability and competitiveness [4,5]. EE1st is therefore more than an accounting rule for emissions. It is a decision principle about which resource is procured first, and in what amount, when demand reduction and supply expansion compete for the same capital.

Most quantitative work on EE1st concerns public decisions in Europe, such as building-sector pathways, power-sector incentives and the design of policy instruments [6–9]. Far less is known about how the principle performs inside individual firms, particularly small and medium-sized enterprises (SMEs) in emerging economies. There, investment decisions are made by owners with limited technical staff, short payback expectations, constrained credit and tariffs that were not designed with efficiency in mind [10–14]. In such firms the dominant "decarbonisation" proposal is often rooftop photovoltaics (PV). PV is visible, financeable and supported by distributed-generation rules, whereas the cheaper but less tangible demand-side measures are postponed or never quantified [15,16].

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Mexico illustrates why this sequencing matters for energy security as well as cost. In 2024 combined-cycle and gas-turbine plants supplied about two-thirds of the electricity injected into the national grid [17]. US pipeline imports covered roughly three-quarters of national natural-gas consumption [18]. The official grid emission factor rose from 0.435 to 0.444 $tCO_2e\ MWh^{-1}$ between 2022 and 2024 [19–21]. Every industrial kWh therefore carries an embedded exposure to imported fossil fuel. The 2025 electricity-sector law also raised the threshold for exempt distributed generation to 0.7 MW [22], which makes on-site PV more attractive for SMEs. For a medium-voltage industrial customer, the question is no longer whether PV pays, but whether efficiency should come first, and what the firm and the grid lose if it does not.

This paper answers that question for a safety-footwear plant in Pachuca, Hidalgo. The work builds on an energy review and measurement-and-verification (M&V) plan prepared during a graduate energy-management project, consistent with ISO 50001 and the national implementation manual [23,24]. We develop a modelling and optimisation framework for this decision. It couples a tariff model built from official charges and validated against the plant's bills, a production-normalised energy baseline and models of each measure with four optimisation problems: the right-sizing of PV after efficiency, efficiency-first pathways that add measures in merit order, a least-capital search over PV and battery designs simulated hourly under three compensation regimes, and a joint cost-optimal design of efficiency, PV and storage formulated as a linear programme in the open energy-system model urbs. Uncertainty is propagated by Monte Carlo simulation, and the hourly PV and storage model is verified against EnergyPlus. All options are compared on common indicators: energy, emissions, cost of conserved energy, net present value (NPV) and a set of plant-level security and resilience metrics. The analysis is an *ex-ante* evaluation study. It does not observe post-retrofit performance, but it is built on measured bills and official tariffs and quantifies uncertainty explicitly, so that its limits are visible. The code is released as an open toolkit that other SMEs can apply to their own bills.

The rest of the paper is organised as follows. Section 2 reviews the literature and states the gap. Section 3 describes the case, data and model. Section 4 presents the results. Section 5 discusses their implications and limitations, and Section 6 concludes.

2. Literature Review

2.1. The energy efficiency first principle

EE1st originates in least-cost and integrated resource planning, in which demand-side resources compete on equal terms with generation and network capacity. Mandel et al. [6] conceptualise it as a decision principle with three ingredients. These are an equal-footing comparison of demand- and supply-side resources, a societal perspective that includes multiple benefits, and a dynamic view of how decisions interact over time. The EU recast Energy Efficiency Directive turned the principle into an obligation for large planning, policy and investment decisions [1]. Commission guidance recommends life-cycle cost-benefit analysis that includes carbon prices and non-energy benefits [2,3]. Below the investment thresholds of the Directive, which include almost all decisions taken by SMEs, the principle is encouraged rather than mandated [3]. Modelling studies show that efficiency-led pathways can lower the total cost of reaching net zero in buildings by avoiding supply and network investment [8]. In the power sector, network-company incentives determine whether demand-side resources are procured at all [7]. Recent work classifies the policy instruments that can operationalise the principle and highlights an implementation gap between legal recognition and actual decisions [9].

Two features of this literature limit its transferability to firms. First, applications are dominated by system-level or sectoral models in high-income economies, where the comparison is between policy pathways rather than between assets that a single owner can buy. Second, the indicators used to judge EE1st, such as system cost and avoided capacity, rarely include the variables that drive industrial decisions. These include tariff

structure, roof-space limits, regulatory risk attached to distributed generation and exposure to energy-price shocks.

2.2. Energy efficiency in industrial SMEs

Industrial SMEs face well-documented barriers: limited information and technical skills, competing investment priorities, split incentives and high perceived risk [10,11]. Energy audits raise awareness, but adoption of recommended measures remains partial. Low-cost organisational measures and measures with short payback are adopted most often [12,15]. Programme design, energy-efficiency networks and access to finance are important levers [13,14,25,26]. Energy management is increasingly understood as a socio-technical practice in which routines, roles and organisational learning matter as much as technology [16]. International uptake of ISO 50001 has been promoted as a scalable route to such practices [27,28]. Non-energy benefits, such as productivity, maintenance, noise and thermal comfort, often equal or exceed the value of the energy saved, yet firms seldom include them in investment calculations [29–31].

For the measures considered here the evidence base is uneven. Compressed-air systems are among the best documented. Leaks commonly waste 20–30% of compressor output in poorly maintained plants, and every 2 psi of excess pressure adds roughly 1% to compressor energy [32–34]. Replacing clutch motors on industrial sewing machines with servo motors is reported to cut motor energy by 65–70% [35]. Idle and base loads frequently dominate specific energy use at low throughput, so operational switch-off strategies can deliver large savings at negligible cost [36–38]. Footwear-specific evidence focuses on product carbon footprints. It shows that factory electricity is a material contributor to the footprint and that results depend strongly on metering resolution [39–41]. Production scheduling has been shown to reduce energy by about 5% [42]. Peer-reviewed quantification of insulation savings for heated barrels and platens is scarce. Practitioner guidance reports short paybacks [43].

2.3. Comparing demand- and supply-side options

The cost of conserved energy (CCE), the annualised cost of a measure divided by the energy it saves, provides a common metric for ranking efficiency against energy prices or supply costs [44,45]. Marginal abatement cost curves extend the idea to emissions. They are criticised for ignoring interactions between measures, uncertainty, non-financial costs and path dependency [46,47]. Supply-side options for industrial SMEs include on-site PV, whose levelised cost has fallen sharply [48]. Mexican case studies report attractive returns under net metering [49,50]. A further option is electrification of process heat, where heat pumps can supply low-temperature heat with coefficients of performance (COP) well above one [51–54]. Because savings are uncertain and energy prices volatile, risk-aware appraisal methods such as Monte Carlo simulation are recommended for efficiency investments [55,56]. Measured savings should be verified ex post through protocols such as IPMVP [57]. Firm-level evidence shows that energy-price uncertainty also shapes manufacturers' investment efficiency [58].

2.4. Optimisation of distributed energy resources

Design optimisation of on-site generation and storage is a mature field. Platforms such as REopt formulate the sizing and dispatch of PV, storage and conventional generation as mixed-integer linear programmes that minimise life-cycle cost under detailed tariffs [59]. For a UK commercial building, integrated optimisation of PV and battery capacity with half-hourly demand and prices found that the combined system depended on ancillary-service revenues and remained a questionable investment, with a payback of about eight years [60]; stochastic programming has been used to design distributed energy systems that remain cost-effective under uncertain demand, prices and solar resource [61]; and, in Mexico City, combining geospatial analysis with energy optimisation showed that rooftop PV in a dense borough could reach about 25% energy autonomy, whereas 90%

required storage [62]. Building performance simulation tools such as EnergyPlus provide physically detailed PV and storage models that are often used to verify simplified models [63], although their predictions depend strongly on input quality; measured rather than tabulated wall transmittances, for instance, changed the predicted heating-plus-cooling demand of a Mexican housing archetype by up to 42% [64]. Demand-side options have been co-optimised with supply in some of these frameworks: DER-CAM has been extended to choose building retrofits together with distributed generation [65], and mixed-integer models have optimised envelope retrofits jointly with conversion and storage units for residential buildings and communities [66,67]. These applications concern building envelopes in residential and commercial settings. In industrial SMEs the electricity demand is still usually an exogenous input: the supply system is optimised for the load as it is, not for the load that cost-effective process measures would leave, and such measures, when included, are typically represented as a fixed reduction rather than as competing resources with their own costs, hourly profiles and uncertainty.

2.5. Research gap and contribution

Taken together, the literature offers a well-defined principle and mature appraisal metrics, but little evidence on how EE1st changes the investment decisions of SMEs under emerging-economy tariffs. Plant-level studies appraise efficiency or PV in isolation, reduce tariffs to an average price, treat security qualitatively and, when optimising PV and storage, take the load as given. This study addresses these gaps with an open, bill-based modelling and optimisation framework, applied to a Mexican footwear SME, that sizes PV and storage after efficiency, co-optimises all three in urbs and reports plant-level security indicators.

3. Materials and Methods

3.1. Case study and data

The plant produces safety footwear in the Pachuca de Soto area, Hidalgo (about 20.1°N, 98.8°W and 2,340 m above sea level). Processes include cutting, stitching, lasting, rubber vulcanising and direct sole injection moulding. Production is flexible. The plant normally operates from 08:00 to 18:00 on weekdays and from 08:00 to 13:00 on Saturdays, and it ran 24-hour shifts during a production surge in November 2025–January 2026. It is supplied at medium voltage by the Federal Electricity Commission (CFE) under the GDMTO tariff (ordinary large demand, medium voltage), Centro Oriente division. Offices and warehouses share the meter but were outside the scope of the audit. The company and its exact location are not disclosed. Its billing, production and audit records are confidential and are reported here only in anonymised and aggregated form. Figure 1 locates the plant at city scale. Figure 2 shows the production sequence, the energy services and electricity share of each area, the supply and compressed-air networks and where each measure acts; it is a schematic, and no architectural drawings or building models are shown. Shares refer to the audited reference month (April 2026).



Figure 1. Location of the plant (Pachuca de Soto, Hidalgo). Boundaries: Natural Earth.

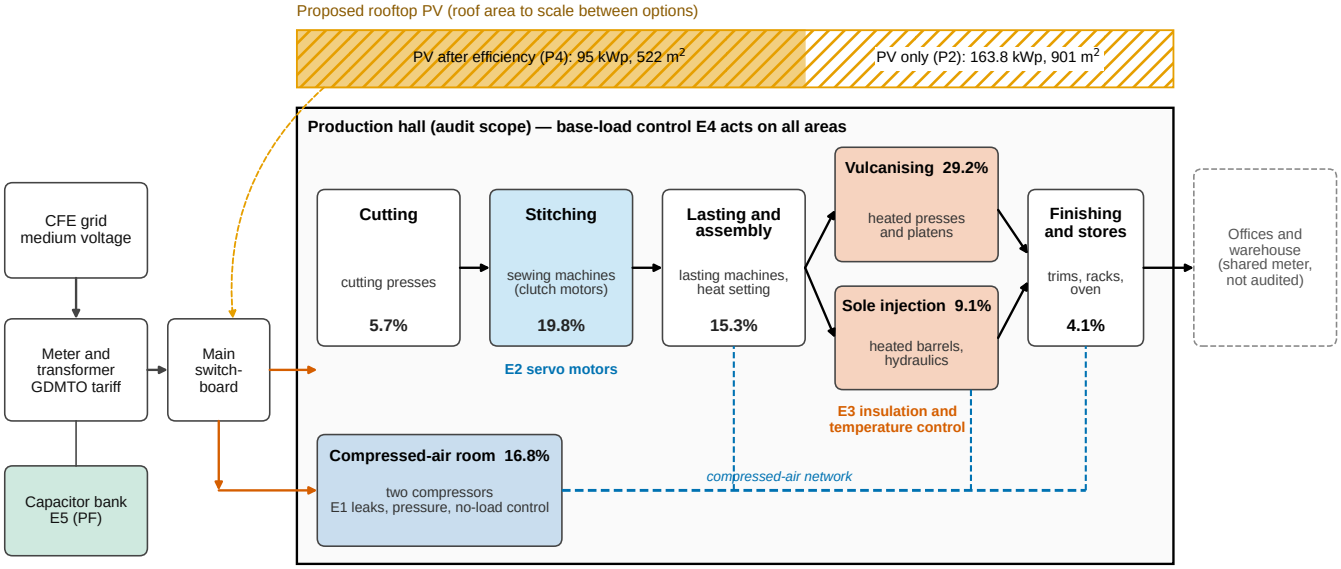


Figure 2. Plant layout and energy flows (schematic, not to scale). Percentages: share of plant electricity; E1–E5: measures.

Table 1 summarises the data and their status. Monthly electricity consumption, billed demand, power factor (PF) and the average price per kWh were transcribed from 28 utility bills (January 2024–April 2026, Table A1). Monthly production in pairs was taken from plant records for 25 of these months. During a walk-through audit in the first half of 2026 the audit team estimated the distribution of electricity use across twelve production areas for a reference month (April 2026). Significant energy uses were vulcanising (29.2%), stitching (19.8%), the compressed-air room (16.8%) and sole injection moulding (9.1%). Tariff charges for every month from January 2024 to October 2026 were retrieved from the utility [68] (Table A2). The emission factor, fuel prices, grid mix and solar resource were taken from official or institutional sources (Table 1). Monetary values are in 2026 Mexican

pesos (MXN; kMXN and MMXN denote thousands and millions) excluding value-added tax (VAT), which firms recover. For reference, 1 USD = 17.10 MXN [69].

Table 1. Data sources and their status.

Item	Content	Source / status
Electricity bills	kWh, billed kW, PF and average price; Jan 2024–Apr 2026 (28 months)	Utility bills, transcribed (measured)
Production	Pairs per month; Apr 2024–Apr 2026 (25 months)	Plant records (measured)
End-use split	Share of 12 areas in the reference month	Audit walk-through estimate
Tariff	GDMTO fixed, energy, distribution and capacity charges, Centro Oriente, monthly 2024–2026	[68,70] (official)
Billed-demand rule	$\min\{D_{\max}, Q/(24\,d\,F_C)\}$, $F_C = 0.55$; PF surcharge and bonus	[70] (official)
Grid emission factor	0.444 $tCO_2e\ MWh^{-1}$ (2024), still mandated for 2026 reporting	[21,71] (official)
Grid mix and gas imports	Gas-fired share of 2024 generation; US pipeline imports vs. consumption	[17,18]
Fuel prices	Natural-gas wholesale index (12-month mean); LPG maximum price for Pachuca	[69,72] (official)
Solar resource	Monthly global and diffuse irradiation, 2001–2025 climatology	[73]
Measure costs	Capital costs of PV, capacitor bank and compressed-air works	Audit quotations; other costs assumed (Table A3)

3.2. Modelling and optimisation framework

Figure 3 summarises the framework. Monthly simulation models of the energy baseline, the tariff, the demand-side measures and rooftop PV (Sections 3.3–3.6) evaluate any portfolio of options. Four optimisation problems, labelled A–D to distinguish them from the portfolios P1–P4, are solved on top of these models. Let $G(\mathcal{E}, k)$ denote annual grid purchases after implementing the set of measures $\mathcal{E}$ and a PV array of k kWp, and G_0 the reference-year purchases.

Problem A, right-sizing PV after efficiency. For the full set of measures $\mathcal{E}^*$, find

$$k^*_{\mathrm{EE}} = \min\{k \geq 0 : G(\mathcal{E}^*, k) \leq G(\varnothing, k_2)\}, \tag{1}$$

where k_2 is the supply-only design. Because G decreases monotonically in k, the problem is solved by bisection to 0.1 kWp.

Problem B, efficiency-first pathways. For each target reduction τ in grid purchases, measures are added in ascending order of CCE until $G \leq (1-\tau)G_0$; if the target is not met, the smallest PV array that closes the gap is found by bisection. The supply-only pathway sizes PV directly from its specific yield for the same τ. Power-factor correction is included in both pathways.

Problem C, least-capital PV and storage design. For a self-sufficiency target σ, find

$$\min_{k \in \mathcal{K},\, b \in \mathcal{B}} C(k, b) \quad \text{subject to} \quad S(k, b) \geq \sigma, \tag{2}$$

where C is capital cost, S is hourly self-sufficiency simulated over 8,760 hours, $\mathcal{K}$ contains 19 PV sizes (0.6–2.4 times the annual net-zero size of each measure set, capped at 700 kWp) and $\mathcal{B}$ 25 battery sizes (0–2,400 kWh; Section 3.9). The 950 designs (two measure sets) are enumerated exhaustively, which guarantees the optimum on the grid without assumptions on the convexity of S, and each is valued under three compensation regimes. Capital cost is minimised, rather than life-cycle cost, because it is the binding constraint for most

SMEs; the NPV of each design is reported alongside. Monte Carlo simulation propagates parameter uncertainty through the monthly models of Problems A and B (Section 3.8).

Problem D, joint cost-optimal design. The fourth problem removes the sequencing assumption altogether. A linear programme built in the open energy-system optimisation model urbs (Section 3.10) chooses simultaneously how much of each efficiency measure to adopt, the PV and battery capacities and the hourly operation of the plant's electricity system, with and without caps on grid emissions. Because it is solved independently of Problems A–C, it tests whether efficiency first emerges from cost minimisation rather than being imposed, and it extends the analysis to deep decarbonisation, where storage becomes necessary.

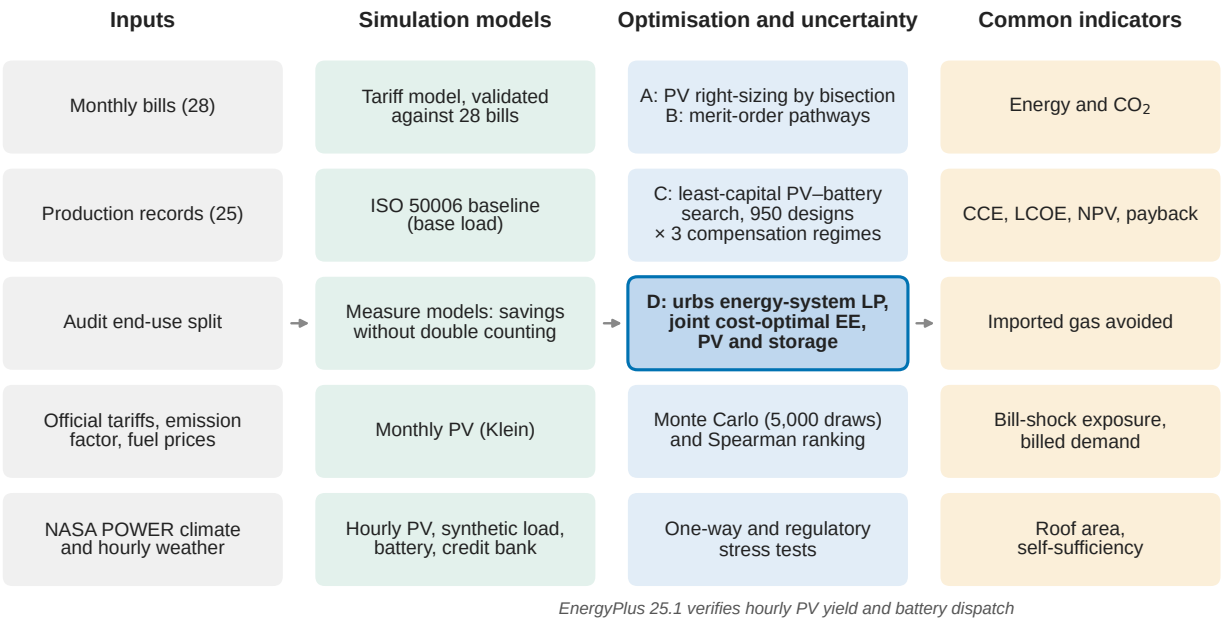


Figure 3. Modelling and optimisation framework.

3.3. Energy baseline and performance indicator

Following ISO 50006 [74], the energy baseline (EnB) relates monthly electricity use E_m to production P_m through ordinary least squares, $E_m = a + b\,P_m + \varepsilon_m$. The intercept a is interpreted as production-independent (base) load. Three specifications were estimated and compared with the regression originally reported by the audit team: all 25 months; 22 months excluding the documented 24-hour-shift months; and the original 16-month subset (April 2024–June 2025 plus September 2025). Model adequacy is reported through R^2 and the coefficient of variation of the root-mean-square error (CV-RMSE). The specification that excludes the 24-hour-shift months is used to define the base load targeted by the operational measure (E4).

3.4. Tariff model and validation

The GDMTO bill (pre-VAT) in month m is

$$B_m = \left[c_f + c_e\,Q_m + (c_d + c_c)\,D_m^b\right](1+\phi_m), \qquad D_m^b = \min\left\{D_m^{\max},\ \frac{Q_m}{24\,d_m\,F_C}\right\}, \tag{3}$$

where c_f is the fixed charge, c_e the energy charge, c_d and c_c the distribution and capacity charges per kW, Q_m the energy consumed, d_m the days in the billing period and $F_C = 0.55$ the load factor prescribed for GDMTO [70]. Because GDMTO has no peak period, the capacity charge always applies the energy-based term; the distribution charge applies the smaller of measured maximum demand and the energy-based term [68]. The PF adjustment

ϕ_m is a surcharge of $\frac{3}{5}(90/\mathrm{PF}-1)$ below 90% PF and a bonus of $\frac{1}{4}(1-90/\mathrm{PF})$ at or above 90%, capped at 120% and 2.5% respectively [70].

The model was validated by applying month-specific official charges to the recorded consumption, billed demand and PF of all 28 months and comparing the result with the average price reported on each bill. The reported billed demand closely matches the energy-based term $Q_m/(24 d_m F_C)$ (mean ratio 1.004). We therefore model billed demand with the energy-based term in all scenarios. Where $Q_m/(24 d_m F_C)$ falls below the measured peak, as appears to be the case here, a kWh saved reduces both the energy charge and the two demand charges. The avoided cost per kWh is then

$$v = c_e + \frac{c_d + c_c}{24\,\bar{d}\,F_C}, \tag{4}$$

evaluated at the October 2026 charges and an average month length $\bar{d} = 30.4$ days. All scenarios use these charges with constant real prices. The reference year is May 2025–April 2026, the latest twelve billed months (460.8 MWh).

3.5. Options appraised

Demand-side measures

Five measures derived from the audit were modelled (Table A3). They are applied sequentially to avoid double counting: E4 first scales all end uses, then component measures act on the remaining energy of their target areas. Sequential application prevents arithmetic double counting but not physical interactions; for example, leak repair (E1) and heaters held at temperature (E3) also contribute to the base load that E4 targets. The saving ranges of Table A3 are wide enough to cover such overlaps, and M&V (Appendix E) is needed to resolve them.

- **E1 Compressed-air package**: leak detection and repair, reduction of discharge pressure and no-load control of the two compressors. The saving is 10–30% of compressed-air energy (central 20%) [32]. Capital cost follows the audit estimate, plus an annual ultrasonic leak survey.
- **E2 Servo motors on sewing machines**: clutch motors in the stitching area are replaced with servo motors that save 50–75% of motor energy (central 65%) [35]. Motors are assumed to use 50–85% of stitching-area energy. The number of machines (63 in the central case) follows from the motor energy, a clutch-motor running power of 0.25–0.45 kW and the scheduled operating hours. It is multiplied by an installed unit cost.
- **E3 Insulation of heated platens and barrels**: insulating jackets and improved temperature control on vulcanising presses and injection-unit barrels. The saving is 10–30% of heater energy (central 20%) [43]. Heaters are assumed to account for 50–80% of the energy of these areas.
- **E4 Base-load (idle) control**: shutdown routines, timers and sub-metering that cut production-independent load by 5–20% (central 10%) [36,37]. The base load is taken as the EnB intercept.
- **E5 Power-factor correction**: the 50 kVAr automatic capacitor bank proposed in the audit, raising PF to 0.95–0.99 (central 0.97). It saves money through the PF adjustment but not energy.

Supply-side options

S1 Rooftop PV is the 163.8 kWp array (252 modules of 650 W; Appendix C) proposed in the audit, connected under the distributed-generation regime with monthly net metering. Monthly plane-of-array irradiation for a south-facing 20° tilt was computed from NASA POWER global and diffuse irradiation using the isotropic-sky monthly method of Klein [75]. Generation applies a performance ratio of 0.75–0.85 [76] and 0.5% per year degradation. Net metering reduces Q_m and therefore the energy-based billed demand. A regulatory

stress case removes this demand credit, so that PV offsets only the energy charge. Monthly surplus, if any, is conservatively assigned zero value; the monthly model does not carry credits between months, whereas the hourly model (Section 3.9) represents the 12-month credit bank.

S2 Fuel substitution and electrification of heat is assessed parametrically, per GJ of useful heat, because the plant's thermal fuel use was not metered during the audit. Five supply routes are compared: liquefied petroleum gas (LPG) or natural-gas boilers (efficiency 0.85), electric resistance, heat pumps (COP 2.5–4.0; [51,52]) and recovery of compressor heat; more than 80% of compressor input power becomes available heat [32], and we conservatively assume that 50–80% of input power (central 70%) is recovered. Prices are the LPG maximum price for Pachuca in October 2026 [72] and the twelve-month mean of the national natural-gas wholesale index [69], doubled to approximate transport and distribution to an end user. Electricity is valued with Eq. (4). Combustion emission factors follow [77]. Because the plant's thermal fuel use is unknown, results are reported per GJ of useful heat and can be scaled by any annual fuel consumption.

3.6. Common indicators

Each option and portfolio is evaluated with the same indicators:

- *Energy*: annual grid purchases and the share avoided.
- *Emissions*: grid purchases multiplied by the official factor of 0.444 $tCO_2e\ MWh^{-1}$, held constant because the official series rose over 2022–2024 [19–21]. A decline of 3% per year would lower lifetime avoided emissions by 24% but would not change any monetary result.
- *Cost of conserved energy*: $CCE = (I \cdot CRF_{r,L} + C_{O\&M})/\Delta E$ [44], where $CRF_{r,L}$ is the capital recovery factor at discount rate r over life L. For PV the analogous metric is the levelised cost of electricity (LCOE). Both are compared with the avoided tariff v.
- *Economics*: NPV over 20 years at a 10% real discount rate, with like-for-like reinvestment of short-lived measures, straight-line residual values and a PV inverter replacement in year 12; simple payback; and the net abatement cost, defined as the annualised negative NPV per tonne of CO_2.
- *Security and resilience*:
  - imported natural gas avoided upstream, $\Delta E_{grid} \cdot s_{gas} \cdot h \cdot s_{imp}$, with gas-fired share of generation $s_{gas} = 0.638$ [17], heat rate $h = 7.2\ GJ\ MWh^{-1}$ and import share $s_{imp} = 0.74$ [18];
  - exposure to an energy-price shock, measured as the increase in the annual bill if the energy charge rose by 50%;
  - the highest energy-implied billed demand, a proxy for proximity to the 100 kW threshold above which GDMTO customers must migrate to the time-of-use GDMTH tariff [68]; PV is not credited with peak reduction because cloudy days and evening operation make its contribution to the measured peak unreliable;
  - the roof area required by PV;
  - the sensitivity of value to a change in distributed-generation compensation.

3.7. Portfolios and efficiency-first pathways

Four portfolios are compared with the baseline (P0): P1, the efficiency bundle (E1–E5); P2, supply only (the 163.8 kWp array); P3, both; and P4, efficiency first, in which the efficiency bundle is implemented and PV is then sized to reach the same grid purchases as P2. To generalise P4, we construct pathways for targets of 5–75% reduction in grid purchases. The supply-only pathway sizes PV directly to the target (with PF correction in both pathways for fairness). The efficiency-first pathway adds measures in ascending order of CCE until the target is met and closes any remaining gap with PV.

3.8. Uncertainty analysis

Parameter uncertainty is propagated with 5,000 Monte Carlo draws from triangular distributions whose bounds are listed in Table A3. These cover measure savings and costs, PV cost, performance and degradation, discount rate (6–14%) and real tariff escalation (-1 to $+2$% per year). The grid-gas parameters do not affect monetary results; their ranges are reported for the imported-gas indicator as low and high bounds. We report 5th–95th percentiles, the probability that each portfolio has positive NPV, the probability that P4 outperforms P2 (with the P4 array both fixed at its central-value size and re-sized in each draw to match P2's grid purchases) and Spearman rank correlations between inputs and the NPV difference P4$-$P2. One-way sensitivities are given in Appendix D. The demand-credit rule for PV is treated as a discrete regulatory scenario rather than a random variable.

3.9. Net-zero scenarios with renewables and storage

Two net-zero definitions are examined because they lead to different designs. *Annual net-zero electricity* requires on-site PV generation to equal the plant's annual consumption, with the grid used as a seasonal buffer. Under the distributed-generation rules, net-metered surpluses are credited in kWh against later bills for up to 12 months, after which unused credits are paid at the local marginal price [78]. We conservatively value expired credits at zero. *Hourly self-sufficiency* measures the share of hourly consumption met on site by PV and a lithium-ion battery, a proxy for physical independence and resilience to supply interruptions. Targets of 70%, 80% and 90% are analysed.

Because interval data are not available, an hourly load profile was synthesised for each reference-year month. It consists of an always-on load of 15 kW (range 8–25 kW, an assumption) and a flat load during scheduled operating hours that closes the monthly energy balance; during the three 24-hour-shift months production is spread over Monday–Saturday. Base-load control (E4) is assumed to reduce the always-on load first. Hourly PV output uses NASA POWER 2023 hourly global and diffuse irradiance at the site [73], transposed to the array plane with an isotropic-sky model and rescaled month by month to the 2001–2025 climatology, which gives a specific yield of 1,523 kWh kWp^{-1}, consistent with the monthly model (1,526 kWh kWp^{-1}). The battery follows a greedy self-consumption dispatch, with 85% round-trip efficiency, a 15-year life and fixed O&M of 2.5% of capital per year [79], 90% usable depth of discharge, a power rating of 0.5 kW per kWh and an installed cost of 8,500 MXN per kWh of nameplate capacity (an assumption anchored on the pack prices of about 200 USD kWh^{-1} reported in [79]; its influence is examined through a break-even analysis).

For each portfolio (supply only, or efficiency bundle first) we sweep PV capacity from 0.6 to 2.4 times the annual net-zero size, capped at the 0.7 MW distributed-generation limit [22], and battery capacity from 0 to 2,400 kWh, and retain the least-capital design that meets each self-sufficiency target. Each design is valued under three compensation regimes: monthly net metering with energy-based billed demand (the central case); net metering in which billed demand follows gross imports rather than net consumption; and net billing, in which imports are billed at the tariff and exports are paid 1.0 MXN kWh^{-1}. PV degradation of 0.5% per year is applied to the PV share of the savings. Battery cost is explored through a break-even analysis; the other parameters of the net-zero module are held at their central values except the always-on load.

The five key designs were cross-validated in EnergyPlus 25.1 [63]. An EPW file was built from the same rescaled NASA POWER hourly data, converted from the NASA POWER time stamps (UTC$-$7 at this longitude) to local clock time; PV was modelled with `Generator:PVWatts` (version 5, 14% system losses, roof-mounted, 20° tilt, south) and a PVWatts inverter (DC/AC ratio 1.1, 96% efficiency); storage with `ElectricLoadCenter:Storage:Simple` using the same capacity, power and efficiencies, AC-coupled and dispatched to track facility demand and store on-site surplus. The same hourly load was imposed through a schedule file, and EnergyPlus hourly purchased and surplus electricity were passed through the same net-metering bank and tariff.

3.10. Joint cost-optimal design with urbs

urbs is an open-source linear optimisation model for distributed energy systems developed at the Technical University of Munich [80]. It represents an energy system as commodities linked by processes and storage, and minimises the annualised cost of meeting an hourly demand. Here the plant is modelled as one site with one electricity bus over one year of 8,760 hours (Figure 4). Grid purchases, rooftop PV, the battery and the four energy-saving measures supply the bus. The plant demand, exports and curtailment draw from it. The hourly load and PV output are those of Section 3.9.

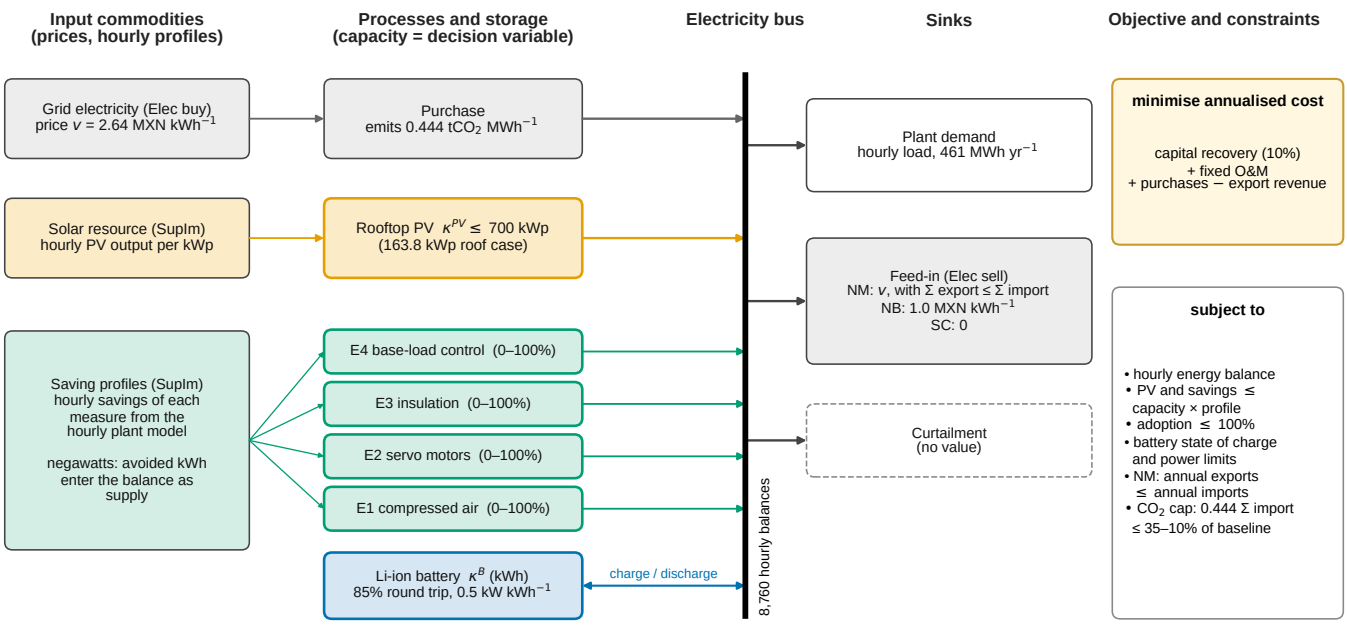


Figure 4. Structure of the urbs model of the plant.

The efficiency measures enter as "negawatt" processes. Let $\hat{s}_{m,t}$ be the hourly saving of measure $m \in \mathcal{M} = \{\mathrm{E1}, \ldots, \mathrm{E4}\}$ at full implementation, taken from the hourly model, and $a_m \in [0,1]$ the adopted fraction. The avoided kWh $a_m \hat{s}_{m,t}$ are supplied to the bus at the measure's investment cost, O&M and lifetime, so that a measure competes with PV and storage as a resource, as the EE1st principle requires. With PV capacity k (kWp) and hourly yield y_t (kWh kWp^{-1}), usable battery capacity b (kWh, 90% of nameplate), grid imports g_t, exports x_t, curtailment c_t, battery charge q_t^+, discharge q_t^- and state of charge e_t, the programme is

$$\min \quad \sum_{m \in \mathcal{M}} a_m(\alpha_m I_m + F_m) + k(\alpha_{\mathrm{PV}} I_{\mathrm{PV}} + F_{\mathrm{PV}}) + b(\alpha_{\mathrm{B}} I_{\mathrm{B}} + F_{\mathrm{B}}) + \sum_{t=1}^{8760} (v\, g_t - p_x\, x_t) \tag{5}$$

$$\text{s.t.} \quad g_t + k\, y_t + \sum_m a_m \hat{s}_{m,t} + q_t^- = d_t + x_t + c_t + q_t^+ \qquad \forall t \tag{6}$$

$$e_t = e_{t-1} + \eta\, q_t^+ - q_t^-/\eta, \qquad 0 \le e_t \le b, \qquad q_t^+,\, q_t^- \le \rho\, b/\delta \qquad \forall t \tag{7}$$

$$0 \le k \le \bar{k}, \qquad 0 \le a_m \le 1 \tag{8}$$

$$\textstyle\sum_t x_t \le \sum_t g_t \quad \text{(net metering only)} \tag{9}$$

$$\varepsilon \textstyle\sum_t g_t \le \phi\, \varepsilon \sum_t d_t \quad \text{(emission cap)}, \tag{10}$$

where d_t is the hourly demand without measures, I and F are investment and fixed O&M costs (for the battery per usable kWh, i.e., the nameplate cost divided by δ), α is the capital recovery factor at the 10% discount rate over each asset's lifetime, $\eta = \sqrt{0.85}$ is the one-way

battery efficiency, ρ the power-to-energy ratio (0.5 kW per kWh of nameplate capacity), $\delta = 0.9$ the usable fraction of nameplate capacity, ε the grid emission factor and ϕ the cap as a fraction of baseline emissions. The battery starts and ends the year half full.

Grid purchases are priced at the marginal value v of Equation (4); this is appropriate because GDMTO demand charges are proportional to energy, whereas the fixed charges and the power-factor adjustment do not depend on the decisions optimised, which is also why power-factor correction (E5) is excluded. The PV investment cost includes the inverter replacement, and the array is limited to $\bar{k} = 700$ kWp, the 0.7 MW distributed-generation limit applied conservatively to DC capacity. Arrays above about 164 kWp exceed the roof area documented in the audit (Section 4.8) and would require additional roof or ground area; the roof-limited scenario bounds this case. Battery capacities from urbs are reported as usable kWh, whereas those of the hourly design search refer to nameplate kWh. The savings of base-load control are assigned to the hours in which it reduces load, preserving its monthly energy. Three compensation regimes, which differ from those of the hourly design search, set the export price p_x: annual net metering, in which exports are credited at v subject to Equation (9); net billing, in which exports are paid 1.0 MXN kWh^{-1}; and self-consumption, in which exports have no value. Surplus PV output can be curtailed. Penalties of 10^{-3} MXN kWh^{-1} on exports and battery throughput remove degenerate solutions with simultaneous import and export or simultaneous charge and discharge; they do not affect the results.

Table 2. Scenarios solved with urbs.

Scenario set	Settings	Runs
Compensation regimes	Net metering, net billing, self-consumption; no emission cap	6
Roof limit	Net metering; $\bar{k} = 163.8$ kWp (audit layout)	2
Emission caps	Net metering; ϕ = 35, 30, 25, 20, 15, 10%	12
Battery cost	Net metering; battery cost 75, 50, 25% of central; no cap and ϕ = 20%	12

Each scenario is solved twice, with the measures available ($a_m \in [0, 1]$) and without them ($a_m = 0$, supply only), which isolates the contribution of efficiency at every point (Table 2). The 32 linear programmes, each with 8,760 hourly time steps, were solved with HiGHS through Pyomo in about 73 s each.

3.11. Reproducibility, data and code

The model is written in Python (NumPy, pandas, Matplotlib) and organised as a small package with scripts that regenerate every number, table and figure in this paper from the input data. All numbers quoted in the text are inserted automatically from the results files. The public repository contains the model code, the official tariff, price, emission-factor and climate series with their sources and access dates, all derived results files, the figures, the Monte Carlo seed and a README with run instructions. The urbs benchmark is generated by two further scripts (input preparation and solution) that run urbs (GPL-3.0, commit 37487e6) with Pyomo 6.8 and HiGHS in a separate Python environment specified in the repository. The plant's billing, production and end-use records are confidential and are not included; the repository documents their format so that the scripts can be run with them, or with another plant's data. Code is released under the MIT licence and public data, results and figures under CC BY 4.0. The repository is archived on Zenodo (https://doi.org/10.5281/zenodo.23419738) and mirrored on GitHub (https://github.com/kerdan85/efficiency-first-sme).

To make the method usable beyond this plant, the same repository contains `ee1st`, a general-purpose implementation installable as a Python package from its `ee1st/` folder [81]. A plant is described in a single case file that lists its bills, tariff, end-use split, candidate measures and PV and storage options; every numeric input can be given as a central value or as a (central, low, high) range with a data-status label (for example measured,

official, literature, quote or assumption). From this file the toolkit produces the core indicators, portfolios, pathways, Monte Carlo statistics and net-zero designs described in Sections 3–4, together with a short report; the EnergyPlus cross-check is not part of it. Three tariff structures are supported: Q-based demand charges as in GDMTO, energy plus measured-peak demand charges as common in Europe and North America, and flat energy prices. Measures are declared as base-load, end-use or power-factor measures, and a production series is optional; without it the base load is supplied directly. The footwear case is included as a configuration file with its public inputs; when the confidential plant records are supplied, an automated test confirms that the toolkit reproduces the deterministic results of this paper exactly and the Monte Carlo statistics within sampling error. An explicitly synthetic example, which runs without any confidential data, shows the inputs for a plant under an energy-plus-demand tariff without production records.

4. Results

4.1. Energy use, baseline and tariff

Monthly consumption ranged from 19,040 to 64,160 kWh (mean 35,523 kWh), and billed demand reached 157 kW (Figure 5a). Power factor averaged 87.1% and fell below the 90% threshold in 25 of 28 months, incurring net PF adjustments of about 13.7 kMXN in the reference year at 2026 charges. Production explains only part of the variability (Figure 5b). Over all 25 months the regression gives $E = 24{,}349 + 0.82\,P$ ($R^2 = 0.29$, CV-RMSE 26.7%). Excluding the 24-hour-shift months yields $E = 25{,}341 + 0.54\,P$ ($R^2 = 0.38$, CV-RMSE 16.0%). The regression in the audit report ($E = 20{,}223 + 0.90\,P$, $R^2 = 0.72$) is reproduced only with the non-contiguous 16-month subset and with one month transcribed as 34,200 instead of 32,400 kWh. With the corrected value the subset gives $R^2 = 0.70$. Whatever the specification, the intercept is large: base load accounts for about 66% of the reference-year mean consumption (95% confidence interval of the intercept 20,601–30,080 kWh per month, or 54–78%). The adopted specification explains little of the variance and does not meet the 15% monthly CV-RMSE criterion of ASHRAE Guideline 14 [82], so the regression is used here to identify the base load, not as an M&V baseline; its uncertainty enters the appraisal through the range of base-load savings. Much of the plant's electricity is therefore not proportional to output, a typical signature of idle equipment, compressed-air leakage and heaters kept at temperature [37,38].

The tariff model reproduces the reported average price of each bill with a mean bias of 2.1% and a mean absolute error of 2.2% (maximum 4.7%; Figure 5c). The small positive bias is consistent with minor charges not itemised in the transcribed data. Two consequences follow. First, the plant's billed demand is set by its monthly energy rather than its instantaneous peak. Under October 2026 charges, an avoided kWh is therefore worth $v = 2.64$ MXN (0.155 USD), of which only 56% is the energy charge itself. Second, billed demand exceeded 100 kW in 5 months, including three consecutive months during the 24-hour shifts. The load factors printed on the bills, if defined on measured maximum demand, imply measured peaks of 121–190 kW, above the 100 kW GDMTO limit in 28 of 28 months. The plant therefore appears to qualify for reclassification to the time-of-use GDMTH tariff, a regulatory exposure discussed in Section 5. The reference-year bill is 1.23 MMXN.

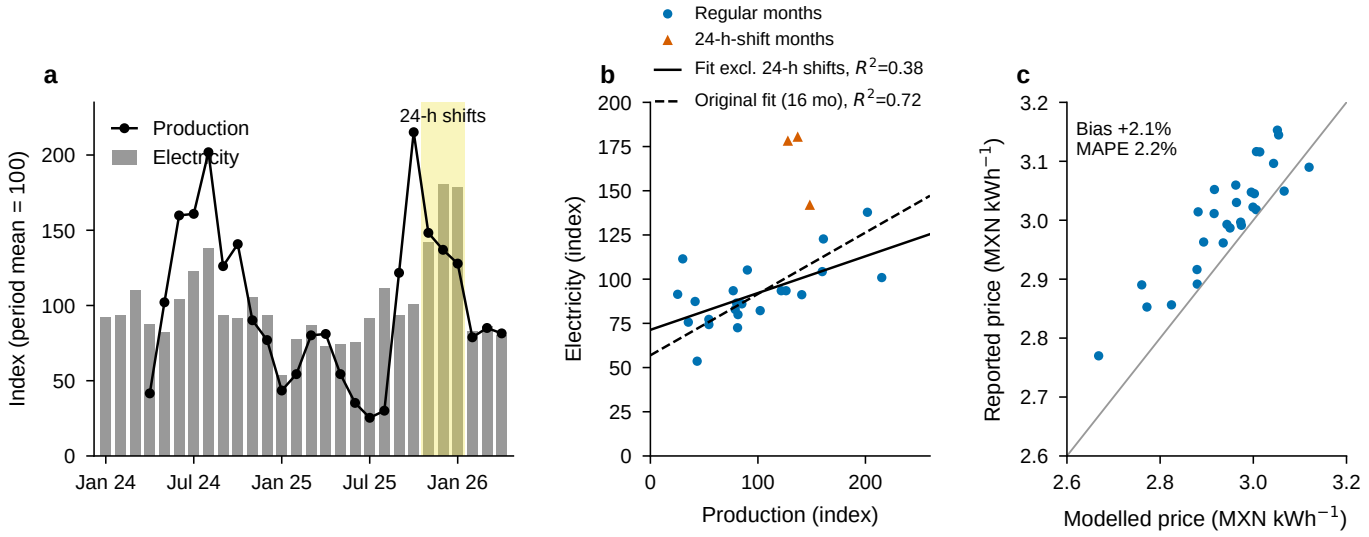


Figure 5. Energy data, baseline and tariff validation: (a) monthly electricity and production (shading: 24-hour shifts); (b) baseline regressions; (c) reported versus modelled price.

4.2. Cost of conserved energy of demand-side measures

Table 3 and Figure 6b rank the measures. Base-load control (E4) is by far the cheapest resource, at 0.21 MXN kWh^{-1} for 30.4 MWh per year. It is followed by insulation of heated platens and barrels (E3, 0.82), servo motors (E2, 0.94, the largest single saving at 41.5 MWh) and the compressed-air package (E1, 1.04). At central values all four lie well below the avoided tariff ($v = 2.64$) and also below the energy charge alone (1.469). Their 5th–95th percentile ranges, from 0.11–0.37 for E4 to 0.78–1.74 for E1, never reach the avoided tariff, and all measures remained below it simultaneously in 99.98% of Monte Carlo draws. Even if only the energy charge were avoided, E1 would still be below it in 85% of draws and the other measures in at least 95%. The measures are of the same order as the PV LCOE (0.92 MXN kWh^{-1}, range 0.74–1.20). E4 was cheaper than PV in 100% of draws and E3 in 65%, whereas E2 (36%) and E1 (25%) were usually slightly more expensive per kWh than PV. The distinctive value of the latter two lies less in their unit cost than in their small capital requirement and their contribution to lower demand (Section 4.5). Power-factor correction saves no energy but removes the PF surcharge and earns the bonus, returning an NPV of 0.21 MMXN on 85 kMXN.

Individually, every measure has a payback of 0.5–2.4 years (Table 3). Combined (P1), they reduce grid purchases by 105.1 MWh per year (23%), avoid 47 tCO_2 per year and require 0.52 MMXN (30 kUSD) for an NPV of 1.84 MMXN.

Table 3. Stand-alone appraisal of each option (CCE: for PV, LCOE; 5th–95th percentiles in brackets).

Option	Capex (kMXN)	Electricity saved (MWh yr^{-1})	CCE / LCOE (MXN kWh^{-1})	Bill saving, yr 1 (kMXN yr^{-1})	NPV (MMXN)	Payback (yr)	CO_2 avoided (t yr^{-1})
E4 Base-load (idle) control	40	30.4	0.21 [0.11–0.37]	82	0.64	0.5	13.5
E3 Insulation of heated platens/barrels	100	23.0	0.82 [0.53–1.41]	61	0.36	1.6	10.2
E2 Servo motors on sewing machines	241	41.5	0.94 [0.73–1.47]	111	0.61	2.2	18.4
E1 Compressed-air package	50	15.5	1.04 [0.78–1.74]	41	0.21	1.5	6.9
E5 Power-factor correction (50 kVAr)	85	0.0	–	36	0.21	2.4	0.0
S1 Rooftop PV (163.8 kWp)	1,802	250.0	0.92 [0.74–1.20]	670	3.57	2.8	111.0

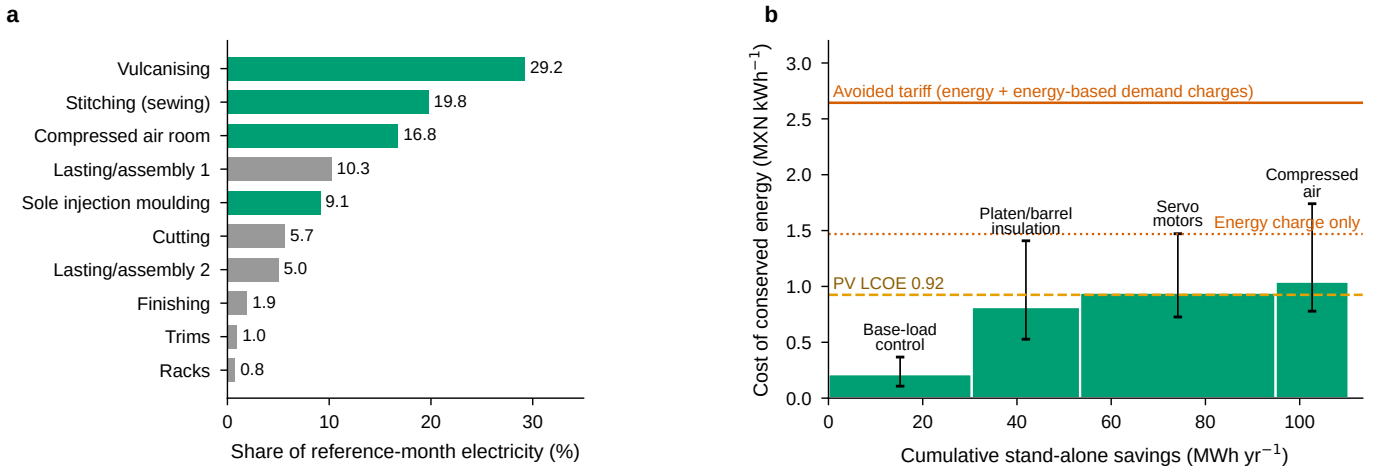


Figure 6. End uses and conservation supply curve: (a) electricity share by area; (b) CCE of each measure (whiskers: 5th–95th percentiles).

4.3. Portfolio comparison

Table 4 and Figure 7 compare the portfolios. Supply-only PV (P2) generates 250 MWh per year with a specific yield of 1,526 kWh kWp^{-1}. It cuts grid purchases by 54% and has a larger absolute NPV (3.57 MMXN) than the efficiency bundle (1.84 MMXN), because it is a larger investment. Per peso invested, however, efficiency returns more: NPV/capex is 16.0 for E4 and between 2.5 and 4.3 for E1–E3, against 2.0 for PV, and P1 beats P2 on this ratio in 99% of Monte Carlo draws.

The efficiency-first portfolio (P4) reaches the same grid purchases as P2 (211 MWh per year, within 0.07 MWh) with 94.9 kWp of PV instead of 163.8 kWp. It requires 13% less capital (1.56 versus 1.80 MMXN) and 42% less roof (522 versus 901 m^2), and achieves a higher NPV (3.80 MMXN, +0.24 MMXN). Emissions fall by 111 t per year in both cases (54% of the baseline 205 t). P4 also lowers the energy-implied billed-demand indicator from 157 to 124 kW. PV is not credited in this indicator (Section 3.6), and because base-load control acts mainly outside production hours, the reduction of the measured daytime peak is likely to be smaller. Installing the efficiency bundle together with the full 163.8 kWp array (P3) cuts grid purchases by 76% and emissions by 155 t per year, with the largest NPV (5.11 MMXN) but also the largest capital requirement (2.32 MMXN). Net abatement costs are negative for every portfolio, between −4,632 and −3,773 MXN tCO$_2^{-1}$.

Table 4. Portfolio comparison on common indicators.

Portfolio	Capex (MMXN)	NPV (MMXN)	Grid purchases (MWh yr^{-1})	Reduction (%)	Residual CO$_2$ (t yr^{-1})	Peak demand (kW)	Shock exposure (kMXN yr^{-1})	Imported gas (GJ yr^{-1})	PV roof (m^2)
P0 Baseline	0.00	0.00	461	0	205	157	342	0	0
P1 Efficiency bundle (E1–E5)	0.52	1.84	356	23	158	124	257	357	0
P2 Supply only: PV 163.8 kWp	1.80	3.57	211	54	94	157	156	850	901
P3 Efficiency bundle + PV 163.8 kWp	2.32	5.11	111	76	49	124	80	1,188	901
P4 Efficiency first + PV 94.9 kWp	1.56	3.80	211	54	94	124	152	850	522

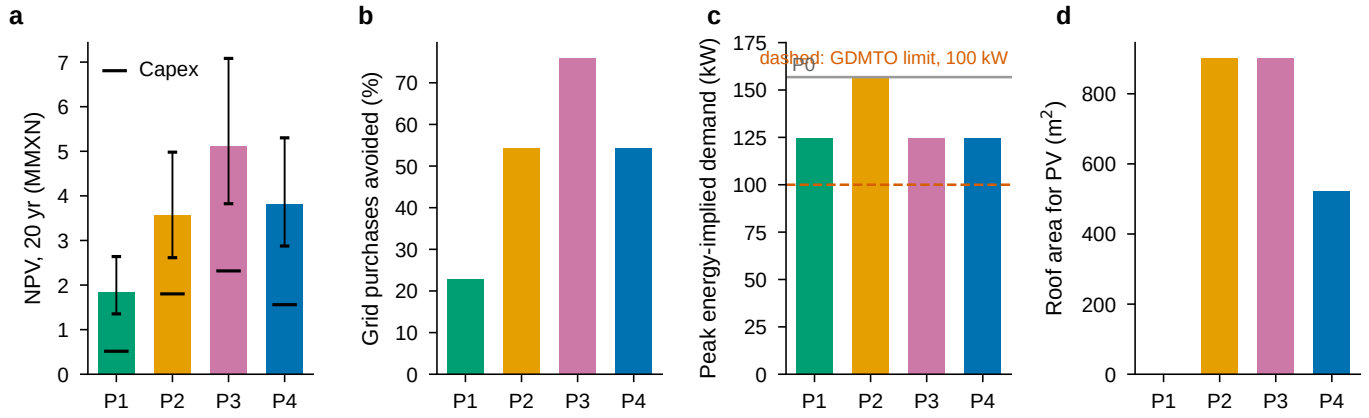


Figure 7. Portfolio indicators: (a) NPV and capital cost; (b) cut in grid purchases; (c) energy-implied billed demand; (d) PV roof area.

4.4. Efficiency-first pathways

Figure 8 generalises the comparison across reduction targets. Up to the 23% reduction delivered by the efficiency bundle, efficiency measures alone meet the target and no PV is needed. Beyond that point PV is still required, but the array is always smaller than in the supply-only pathway. At a 50% target, for instance, the efficiency-first pathway needs 82 kWp instead of 151 kWp, 1.42 instead of 1.75 MMXN of capital (19% less) and 451 instead of 830 m^2 of roof, with an NPV gain of 0.21 MMXN. The NPV gain is positive for every target. It is largest where efficiency alone meets the target and remains roughly constant at intermediate targets, where both pathways add PV at the same marginal value. It rises again above about 65%, where the supply-only array starts to produce monthly surpluses that net metering does not remunerate in this model. The steps at low targets arise because measures are added as discrete units in merit order and can exceed the target.

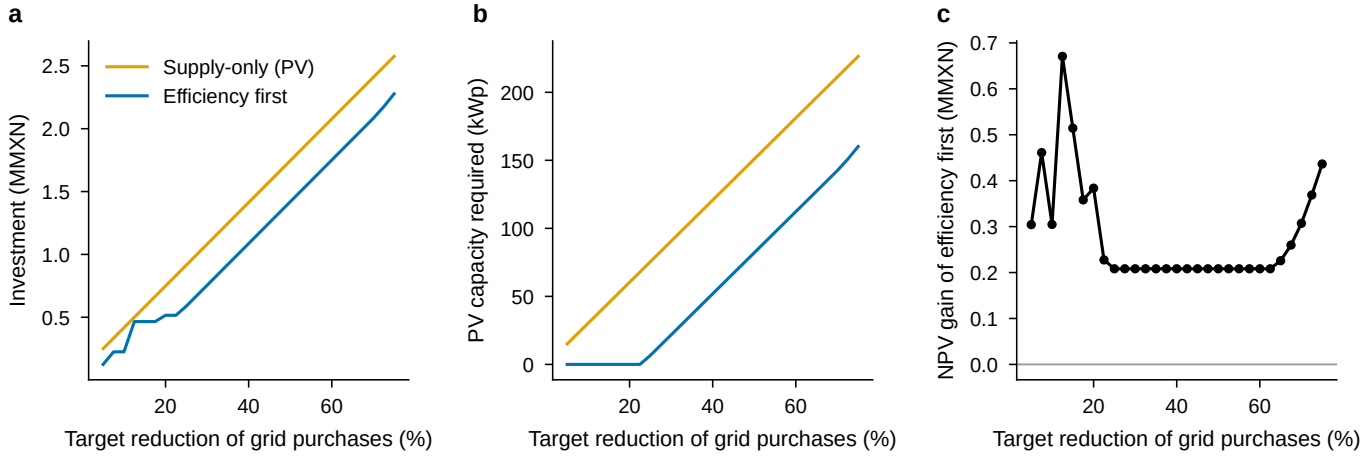


Figure 8. Supply-only versus efficiency-first pathways: (a) capital; (b) PV capacity; (c) NPV gain of efficiency first.

4.5. Energy security and resilience

The security indicators favour efficiency in ways that the averaged economics do not reveal (Figure 9).

- *Regulatory stress.* If distributed-generation output no longer reduced billed demand, for example under a move from net metering to net billing, the NPV of the PV array would fall by 69% to 1.11 MMXN. The NPV of the efficiency bundle would be unchanged (1.84 MMXN), because its savings reduce consumption behind the meter.

- *Energy-price shock.* A 50% increase in the energy charge would raise the baseline bill by 342 kMXN per year. The efficiency bundle cuts this exposure by 25%, and P4 cuts it by 56%, at least as much as P2 but with less capital.
- *Imported gas.* Upstream, the grid electricity avoided by P1 corresponds to about 357 GJ per year of imported natural gas (287–474 GJ across the parameter ranges), and that avoided by P2 and P4 to 850 GJ (683–1,127 GJ).
- *Tariff-class threshold.* Efficiency reduces the highest energy-implied billed demand by 32 kW. Neither this indicator in the surge months nor the measured peaks implied by the bills fall below the 100 kW GDMTO limit, so peak management (scheduling of compressors, presses and heaters) would be needed if the plant is reclassified.

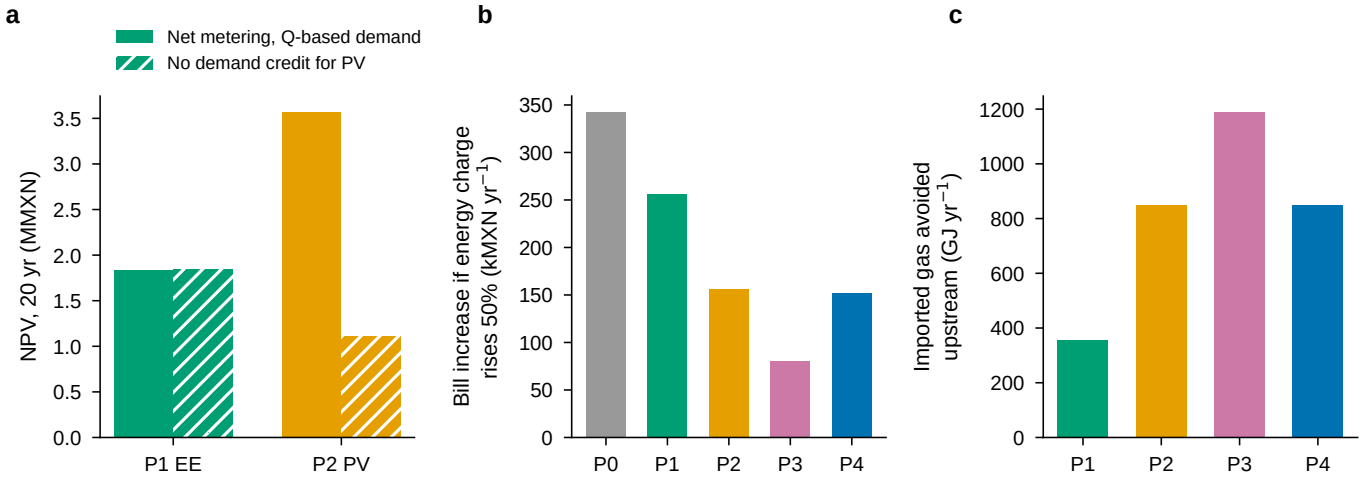


Figure 9. Resilience indicators: (a) NPV with and without PV demand credit; (b) bill increase if the energy charge rises 50%; (c) imported gas avoided.

4.6. Fuel substitution and electrification of heat

Per GJ of useful heat, an LPG boiler costs 424 MXN and emits 74 kgCO_2. A natural-gas boiler costs 126 MXN and emits 66 kgCO_2 (Figure 10). Electric resistance heating at the avoided tariff costs 734 MXN GJ^{-1} and emits 123 kgCO_2 GJ^{-1}, more than either fuel on both counts. A heat pump with COP 3 costs 245 MXN and emits 41 kgCO_2 per GJ. It beats LPG on cost for any COP above 1.7 and on emissions above 1.7. Against pipeline natural gas, however, a heat pump breaks even on emissions at COP 1.9 but on cost only at COP 5.8, because wholesale gas is cheap relative to GDMTO electricity. Electrification is therefore a strong option where the plant burns LPG and a weak one where pipeline gas is available, unless electricity prices or gas prices change.

Efficiency interacts with electrification in two ways. First, it creates headroom: the 105.1 MWh saved each year by P1 could supply about 1,135 GJ of useful heat through a COP-3 heat pump without increasing grid purchases above the baseline. Second, it reduces the heat available for recovery. Compressor heat that could be recovered falls from about 195 to 146 GJ per year once E1 and E4 are implemented. Recovery projects should therefore be sized after the compressed-air system has been optimised, not before.

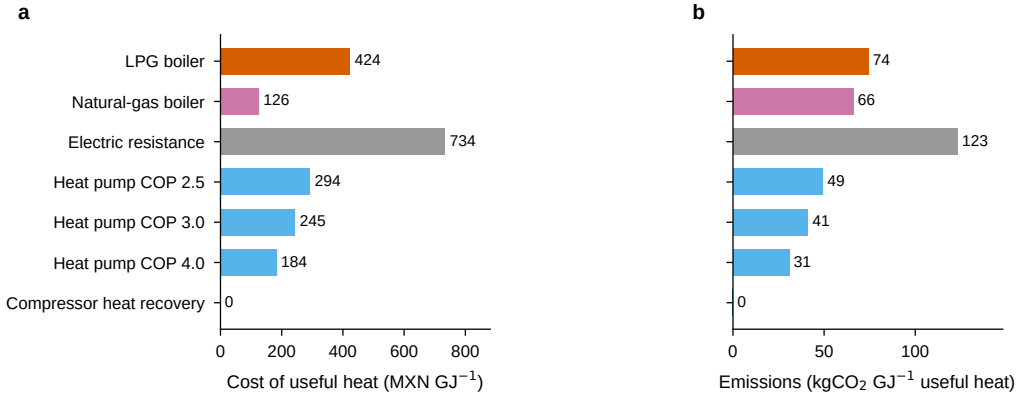


Figure 10. Heat supply routes: (a) cost and (b) emissions per GJ of useful heat.

4.7. Uncertainty

All portfolios had positive NPV in every Monte Carlo draw (Figure 11a). The 5th–95th percentile ranges were 1.35–2.64 MMXN for P1, 2.61–4.98 for P2 and 2.88–5.30 for P4. P4 outperformed P2 in 86% of draws. In these draws the P4 array is held at its central-value size, so P4 reaches at least P2's cut in grid purchases in only 57% of them. When the array is instead re-sized in every draw to match P2's grid purchases (81–105 kWp, 5th–95th percentiles), efficiency first requires less capital in 99% and achieves a higher NPV in 97% of draws. The advantage of efficiency first is driven mainly by the achievable base-load saving ($\rho = 0.67$), followed by the insulation saving ($\rho = 0.34$) and, at similar magnitude, the servo saving ($\rho = 0.27$) and PV cost ($\rho = 0.27$). It shrinks when PV performs better than expected ($\rho = -0.21$; Figure 11b). One-way sensitivities (Appendix D) confirm that no single parameter within its range reverses the sign of any portfolio NPV.

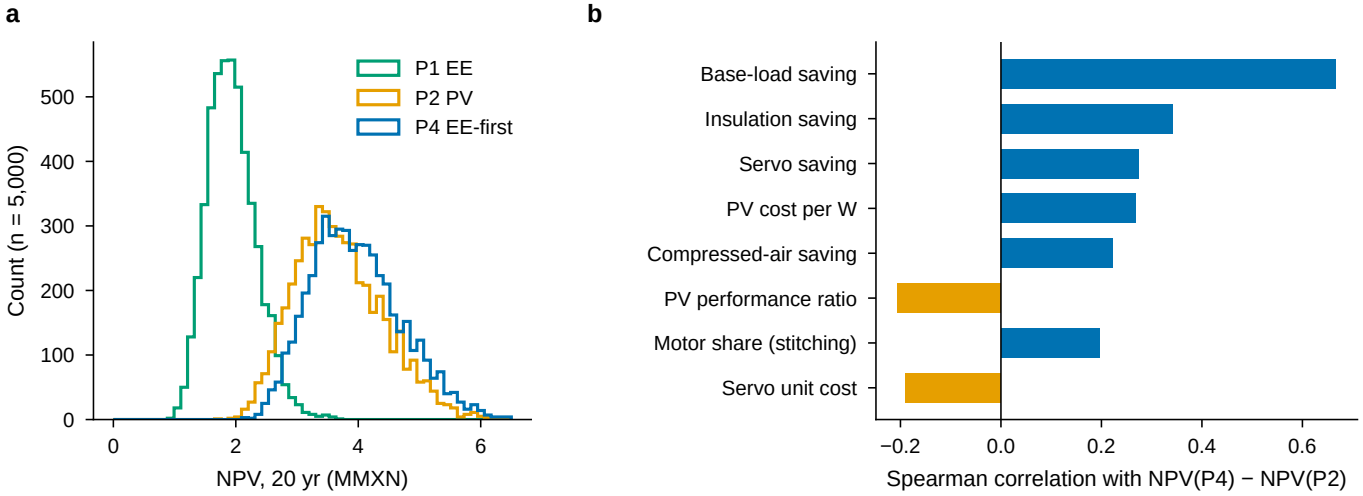


Figure 11. Uncertainty (5,000 draws): (a) NPV distributions; (b) main drivers of the NPV difference P4 − P2.

4.8. Pathways to net-zero electricity

Annual net-zero electricity is technically and economically within reach (Figure 12). Without efficiency, the plant needs 303 kWp of PV on 1,665 m^2 of roof (3.41 MMXN). With the efficiency bundle first it needs 234 kWp on 1,285 m^2 (3.09 MMXN, including the measures), that is 69 kWp and 380 m^2 less, and its NPV is higher (6.65 versus 6.44 MMXN). Under net metering with a 12-month credit bank the annual bill falls to the fixed charge of about 4.0 kMXN, and no credits expire, because the seasonal surplus of the dry months

is used during the high-consumption winter months (Figure 12c). Both designs stay well below the 0.7 MW distributed-generation limit. Roof space is the binding physical constraint. The usable roof area is not documented; the audit's 252-module layout implies at least about 901 m^2. On that area the full efficiency bundle plus the 163.8 kWp array (P3) cuts grid purchases by 76%, against 54% for the array alone, whereas annual net zero would need about 380 m^2 more roof without efficiency than with it. The less roof a plant has, the more efficiency first determines how close it can get to net zero.

Annual net zero is not physical independence. Without storage only 57% of hourly consumption is met on site, and gross grid imports remain 152 MWh per year in the efficiency-first design (67 tCO_2 on a location basis) and 199 MWh in the supply-only design. Self-sufficiency above the roughly 57% reached without storage requires batteries. Beyond about 57% (efficiency first) and 70% (supply only) self-sufficiency, the least-capital designs also oversize the array, and the cost rises steeply (Figure 12a). The least-capital efficiency-first design for 80% self-sufficiency combines 350 kWp with 300 kWh (6.92 MMXN, NPV 1.80 MMXN); the supply-only equivalent needs 424 kWp and 400 kWh, 1.23 MMXN more, and has an NPV of only 0.39 MMXN, which turns negative if billed demand follows gross imports (−0.47 MMXN). At 90% the efficiency-first design needs 467 kWp and 800 kWh (12.46 MMXN) and its NPV turns negative (−5.35 MMXN). The highest self-sufficiency within the range explored (PV up to 2.4 times the annual net-zero array or 0.7 MW, and up to 2,400 kWh of storage) is about 98%, with 561 kWp and 2,400 kWh. Efficiency first lowers the capital required at every self-sufficiency level. The advantage comes mainly from lower total consumption, which shrinks the array: a supply-only design given the lower night load of the efficiency-first plant still needs 8.15 MMXN for 80% self-sufficiency.

Storage does not pay under the current rules. Under net metering the grid already acts as a free, lossless seasonal store, so adding 200 kWh to the efficiency-first net-zero design raises self-sufficiency by 14 percentage points but changes NPV by −2.49 MMXN. The loss shrinks if billed demand follows gross imports (−1.93 MMXN) or under net billing (−1.72 MMXN), but remains negative at current battery costs (Figure 12b). Storage would break even only with battery costs 75% lower under net billing or 85% lower under import-based demand charges, and never under net metering, where round-trip losses add to net imports. The value of storage for this plant is therefore resilience: the 80% design holds about 27 hours of the always-on load. These results depend on the synthetic profile. Self-sufficiency of the annual net-zero designs ranges from 51% to 60% across the assumed always-on loads.

The EnergyPlus runs verify the PV yield and the battery dispatch of the Python model; they use the same synthetic load, weather, credit bank and tariff, so they do not test those inputs (Table 5). For the designs with storage, self-sufficiency differs by at most 0.5 percentage points and grid imports by at most 5.5%; for the annual net-zero designs without storage the difference is at most 0.2 points. EnergyPlus DC output is within 0.2% of the Python AC output, and its AC yield (1,456 kWh kWp^{-1}) is −4.4% lower mainly because the PVWatts inverter efficiency (96%) is applied on top of 14% system losses, whereas the Python model uses a single performance ratio of 0.80. Sized with the EnergyPlus yield, annual net zero requires 244 kWp with efficiency first and 316 kWp without it (a gap of 72 kWp), and the NPVs become 6.52 and 6.27 MMXN, so the ranking is unchanged.

Table 5. Net-zero designs: Python model versus EnergyPlus 25.1 (annual values).

Design	PV (kWp)	Battery (kWh)	PV output (MWh)	Grid imports (MWh)	Exports (MWh)	Self-sufficiency (%)
NZ-S annual net zero, supply only	302.7	0	461 / 441	199 / 200	199 / 180	56.8 / 56.6
NZ-E annual net zero, efficiency first	233.6	0	356 / 340	152 / 153	152 / 137	57.3 / 57.1
80% self-sufficiency, supply only	423.7	400	645 / 617	91 / 89	260 / 229	80.3 / 80.7
80% self-sufficiency, efficiency first	350.4	300	534 / 510	71 / 70	238 / 213	80.0 / 80.4
90% self-sufficiency, efficiency first	467.2	800	711 / 680	35 / 33	374 / 341	90.2 / 90.7
NZ-S re-sized with EnergyPlus yield	316.5	0	– / 461	– / 197	– / 197	– / 57.2
NZ-E re-sized with EnergyPlus yield	244.3	0	– / 356	– / 150	– / 150	– / 57.7

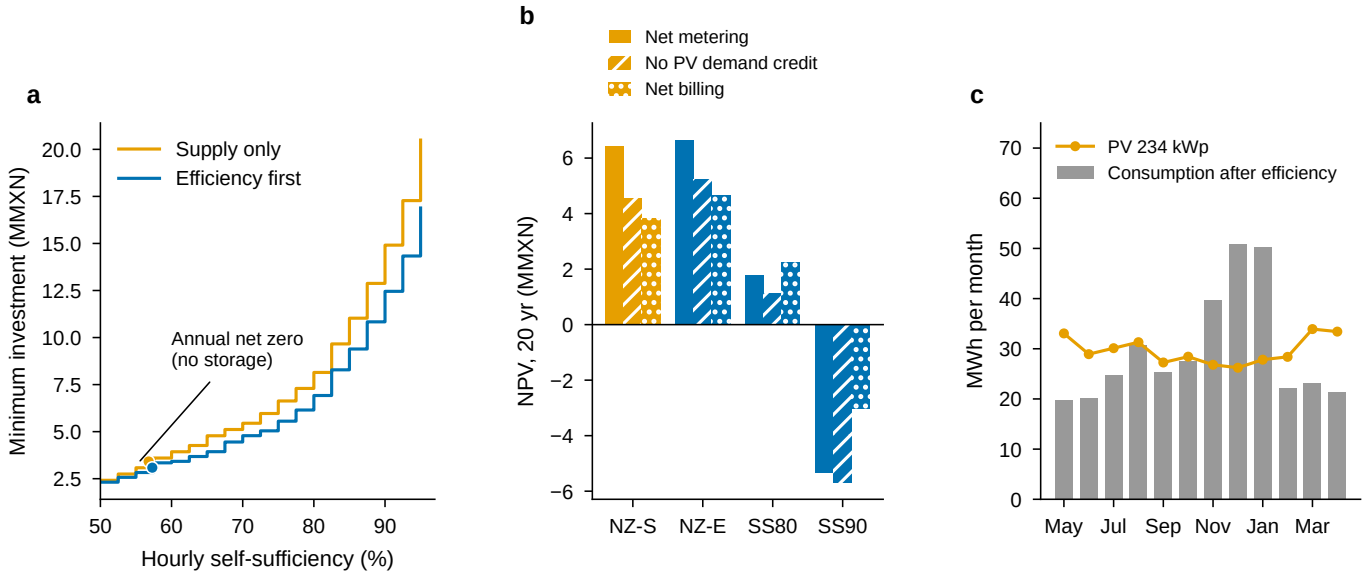


Figure 12. Net zero with PV and storage: (a) least capital by self-sufficiency target; (b) NPV under three compensation regimes; (c) monthly balance of the efficiency-first net-zero design.

4.9. Joint cost-optimal design with urbs

4.9.1. Cost-optimal designs under three compensation regimes

Without an emission cap and under annual net metering, the cost-optimal supply-only design is a 303 kWp array without storage (Table 6). This is the annual net-zero size found by the hourly design search in Section 4.8, with the same hourly self-sufficiency of 57%, so the two models describe the same system. When efficiency is available, the optimiser adopts base-load control and insulation in full but not the servo motors or the compressed-air package. It installs 269 kWp of PV and lowers the annualised cost from 412 to 391 kMXN (5%). This choice follows the merit order of Section 4.2. Under annual net metering every kWh of PV output is worth the full tariff, so a measure enters only if its CCE is below the PV LCOE, and E2 and E1 lie just above it.

As soon as PV output loses part of its value, all four measures enter the optimum. With PV limited to the 163.8 kWp array, without export compensation, or with net billing at 1.0 MXN kWh^{-1}, efficiency lowers the annualised cost by 25%, 14% and 16%, respectively. Under net billing the export price exceeds the PV LCOE, so the array grows to the 0.7 MW limit with or without efficiency. The similar PV sizes and imports of the self-consumption and roof-limited designs with efficiency are a coincidence of the two constraints. In no regime are batteries part of the cost-optimal design without an emission cap.

Table 6. Cost-optimal designs from urbs without an emission cap (grid only: 1,218 kMXN yr^{-1}).

Compensation regime	Efficiency	Measures adopted	PV (kWp)	Battery (usable kWh)	Grid imports (MWh)	CO_2 (t)	Cost (kMXN yr^{-1})
Net metering	with	E3, E4	269	0	172	76	391
	without	–	303	0	199	88	412
Net billing (1.0 MXN kWh^{-1})	with	E1–E4	700	0	118	53	518
	without	–	700	0	162	72	615
Self-consumption only	with	E1–E4	164	0	177	79	772
	without	–	208	0	232	103	897
Net metering, PV $\leq$ 163.8 kWp	with	E1–E4	164	0	177	79	585
	without	–	164	0	261	116	782

4.9.2. Decarbonisation frontier and order of adoption

Caps on gross grid emissions trace the cost-optimal decarbonisation frontier (Figure 13). The unconstrained optima already reduce gross grid CO_2 to 37% of the baseline with efficiency and 43% without it. As the cap tightens, the optimiser adopts resources in a clear order (Figure 13c). It first completes the efficiency bundle: at a 35% cap the servo

motors are adopted at 73%, and all measures are adopted from 30% onwards. Storage is added next, with the array re-optimised slightly downwards, and PV grows beyond its unconstrained optimum only at caps of 20% or less. Batteries enter at a cap of 35% without efficiency but only at 30% with it. At a 20% cap the efficiency-first optimum needs 285 kWp of PV and 163 kWh of storage instead of 452 kWp and 317 kWh. At 10% it needs 441 kWp and 536 kWh instead of 625 kWp and 826 kWh.

The cost of the frontier rises steeply once storage is required (Figure 13a). With efficiency, the annualised cost is 455 kMXN at a 30% cap and 709 kMXN at 20%, against 703 and 1,083 kMXN without it. Across the caps examined, efficiency lowers the annualised cost of meeting the cap by 29–35%. At a 10% cap the annualised cost is 1.47 MMXN with efficiency and 2.07 MMXN without, both above the 1,218 kMXN per year of buying all electricity from the grid. Net grid purchases are close to zero along the frontier, because annual exports offset imports under net metering. Storage then accounts for 54% of the annualised cost with efficiency and 59% without it, whereas the measures account for only 5% (Figure 13d).

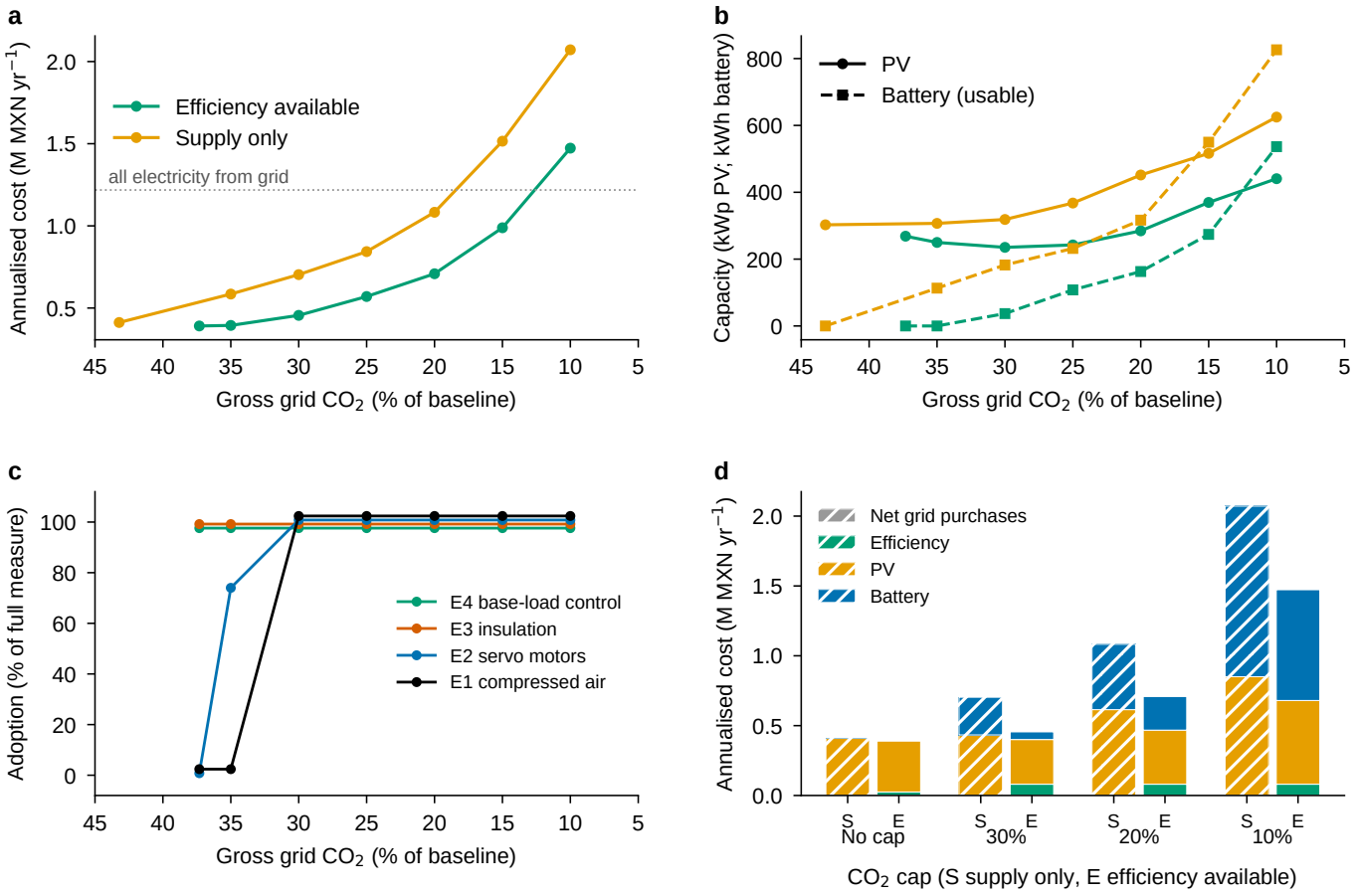


Figure 13. Cost-optimal decarbonisation frontier from urbs (net metering): (a) annualised cost; (b) PV and battery capacity; (c) adoption of each measure; (d) annualised cost by component.

4.9.3. Hourly operation

Figure 14 shows how the two optimal systems for a 20% cap operate in a dry-season week and in a week of the 24-hour shift months. The negawatts of the measures lower the load in every hour, including the nights and weekends that PV cannot reach. In the efficiency-first system the battery shifts midday PV surplus to the evening and night; 59% of its discharge occurs in hours without PV output, and it performs about 306 equivalent full cycles per year (267 in the supply-only system). In the dry season, exports are limited

by the annual net-metering constraint, so part of the midday surplus is curtailed: 16% of annual PV output with efficiency and 31% without it. In the shift weeks, round-the-clock production raises night demand, the battery is emptied every night and the grid covers the remainder. Both systems import the same 92 MWh allowed by the cap. Because efficiency lowers consumption in every hour, the efficiency-first system meets the cap with a smaller array, half the storage and much less curtailment.

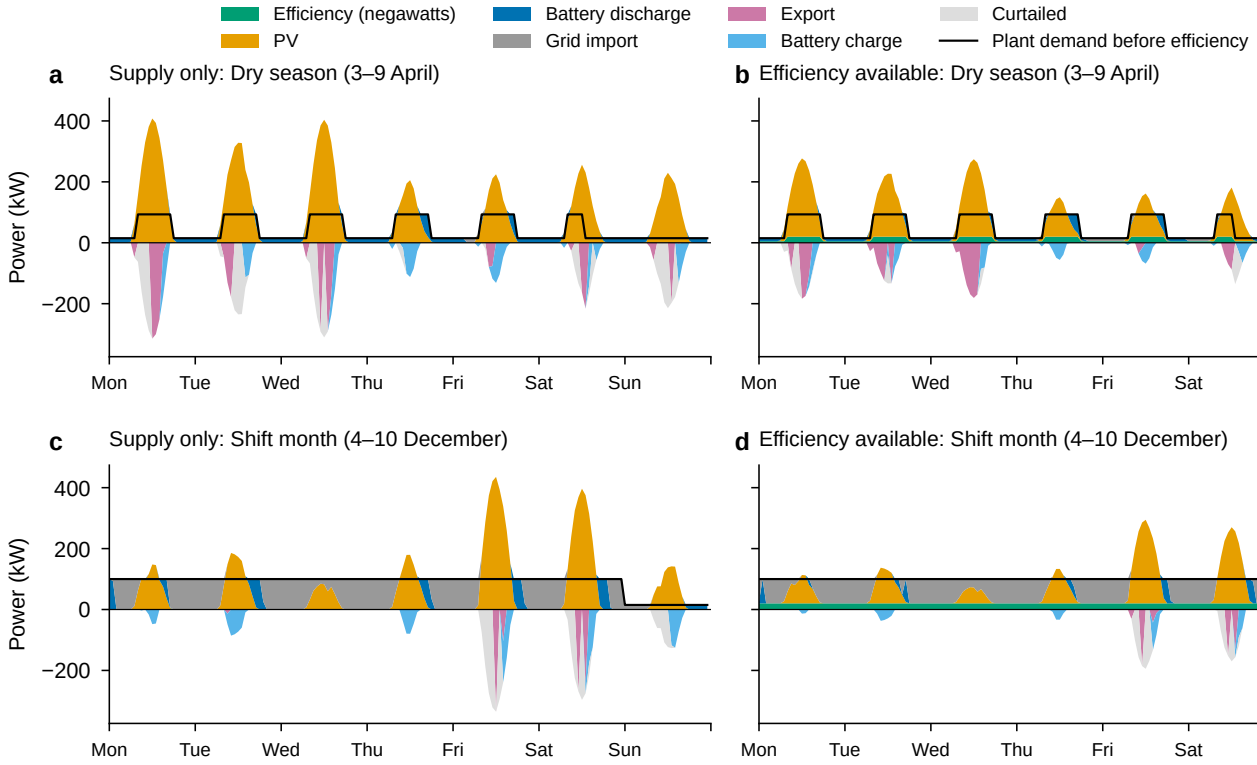


Figure 14. Hourly operation of the cost-optimal systems for a 20% emission cap: (a,c) supply only; (b,d) efficiency available.

4.9.4. Battery cost

Table 7 repeats the optimisation with lower battery costs. Without an emission cap, batteries do not enter the optimum even at a quarter of the central cost, because under annual net metering the grid acts as a free and lossless store. Under a 20% cap, halving the battery cost changes the optimal storage to 189 kWh with efficiency and 366 kWh without it, and at a quarter of the cost to 211 and 432 kWh. Even then, efficiency lowers the annualised cost by 28%. Cheaper storage therefore does not displace efficiency; efficiency reduces both the storage required and the cost of every step along the frontier.

Table 7. Effect of battery cost on the urbs optimum under net metering (battery in usable kWh).

	No emission cap						CO_2 cap 20% of baseline					
	Supply only			Efficiency available			Supply only			Efficiency available		
Battery cost	kWh	kWp	kMXN	kWh	kWp	kMXN	kWh	kWp	kMXN	kWh	kWp	kMXN
100%	0	303	412	0	269	391	317	452	1,083	163	285	709
75%	0	303	412	0	269	391	341	428	980	172	276	657
50%	0	303	412	0	269	391	366	409	871	189	264	601
25%	0	303	412	0	269	391	432	375	747	211	252	539

5. Discussion

5.1. What efficiency first means at plant scale

The results do not support a simple "efficiency instead of solar" narrative. Rooftop PV is a profitable investment for this plant, and the largest NPV is obtained by doing both. Under the current net-metering rules NPV keeps rising with PV capacity up to the annual net-zero size, so a firm with ample capital and roof area would build the larger array together with the efficiency measures. P4 is the better choice when capital, roof area or connection capacity is limited, which is the usual situation of SMEs: among the designs that reach at least P2's cut in grid purchases, it earns the most NPV per MXN invested (2.44, against 1.98 for P2, 2.21 for P3 and 2.15 for the efficiency-first net-zero design). The efficiency bundle alone (P1) earns 3.57 per MXN but stops at a 23% cut. What EE1st changes is therefore the order and size of the supply investment. Treating demand-side measures as a resource that competes with PV, as the principle requires [2,6], reveals that efficiency delivers the first 23% of the reduction in grid purchases at low cost. It also shrinks the PV array needed for any higher target, by 69 kWp in this case. That frees capital, roof area and grid-connection capacity for later steps such as heat electrification. The same logic underpins system-level EE1st analyses [7,8]. Here it appears in the investment decision of a single SME.

The low cost of the operational measure (E4) deserves emphasis. Its value rests on the large production-independent load revealed by a simple baseline regression. Such loads are common in discrete manufacturing [36,37] but invisible in averaged indicators such as kWh per pair. Capturing these savings requires routines, sub-metering and accountability rather than capital, which is why energy management practices and ISO 50001 matter for SMEs [16,27]. Efficiency measures can also bring non-energy benefits, such as lower noise from compressed-air systems [31]. Insulating heated equipment would reduce heat gains in a plant where workers reported high indoor temperatures. These benefits are not monetised here, so the estimates for efficiency are conservative [29,30].

5.2. Tariff design as a hidden driver

The tariff analysis explains why both efficiency and PV look attractive. Under GDMTO the capacity charge, and in practice also the distribution charge, is computed from monthly energy divided by a fixed load factor [70]. This turns demand charges into a volumetric charge and raises the value of an avoided kWh to 2.64 MXN, 56% of which is the energy charge. For efficiency the design is benign, since saved kWh are worth the same whatever their timing. For PV it creates a regulatory dependency, because most of its value rests on net-metered generation reducing the billed demand. The stress case shows that a change in that rule would remove most of the PV's NPV while leaving efficiency untouched. An EE1st appraisal that uses only an average price would miss this asymmetry. The finding echoes the importance of incentive design stressed by Pato et al. [7], seen here from the customer's side of the meter. The bills also suggest that the plant's measured peaks exceed the 100 kW GDMTO limit. Under the time-of-use GDMTH tariff, capacity is billed on demand coincident with the evening peak period and energy is priced by period [70]. Base-load reductions that persist into the peak period would then gain value, whereas daytime PV output would contribute little to peak-period demand. Reclassification would therefore be likely to widen the advantage of efficiency first, although this was not simulated.

5.3. Energy security and the phase-out of fossil fuels

Because nearly two-thirds of Mexican grid electricity is gas-fired and roughly three-quarters of the gas is imported [17,18], every kWh saved or self-generated by an SME reduces the country's exposure to an external fuel supply. At the plant level, efficiency and PV reduce exposure to an energy-charge shock by similar amounts for the same reduction in grid purchases. Efficiency, however, does so without regulatory risk, without roof constraints and while lowering billed demand. For the phase-out of fossil fuels, the thermal analysis adds a nuance. Electrification of heat is economically compelling against

LPG but not yet against pipeline gas. Efficiency first makes room for electrification by freeing grid capacity and lowering the plant's energy-implied billed demand. The official emission factor has not declined in recent years. Behind-the-meter efficiency is therefore the decarbonisation lever least dependent on progress in the grid.

5.4. Can the plant reach net zero?

For electricity, yes, on an annual basis and at a profit under current rules: an efficiency-first plant with about 234 kWp of rooftop PV generates as much as it consumes. Two qualifications matter for how such a claim is made. First, annual netting is not equivalent to zero emissions in a location-based inventory: about 67 tCO_2 per year of grid imports remain, mostly at night and in the winter shift months, even though exports displace a similar amount of grid generation during the day. Second, physical self-sufficiency above about 57% requires batteries whose incremental cost is not recovered under any of the three compensation regimes tested (net metering with energy-based or with import-based demand charges, and net billing). A net-zero claim for this plant should therefore be framed as annual net-zero electricity, with gross imports and location-based emissions reported alongside [23]. Under the GDMTO tariff, storage becomes attractive only if battery costs fall by at least three-quarters, if compensation rules penalise exports more than in the cases tested, or if the firm values resilience to outages explicitly. Under the time-of-use GDMTH tariff, which the plant's measured peaks may eventually trigger, storage could also earn peak-period capacity and energy value; this case was not modelled. Efficiency first reduces the cost of every one of these steps, chiefly by shrinking the consumption that PV and storage must cover, and with it the array and the roof. Full net zero also requires the thermal fuels discussed in Section 4.6 to be electrified, which the electricity headroom created by efficiency makes easier.

5.5. What formal optimisation adds

The urbs results support the sequencing rule and refine it. In the linear programme, efficiency is not placed first by assumption: it competes with PV, storage and grid purchases in every hour, and the optimiser chooses it first. Its order of adoption reproduces the conservation supply curve. Base-load control and insulation are adopted in every scenario, and the servo motors and the compressed-air package are adopted as soon as the value of PV output falls below the tariff. Under generous annual net metering, PV output is worth the full tariff, so the two measures whose CCE is slightly above the PV LCOE are not selected even though each is profitable on its own (Table 3); the ranking between PV and these two measures is also uncertain in the Monte Carlo analysis. When roof area, export compensation or an emission cap binds, the efficiency bundle enters first (in full from a 30% cap) and lowers the cost of every subsequent step. These conditions characterise the deeper stages of decarbonisation, and the steps include the storage that physical self-sufficiency requires. Efficiency first is therefore an emergent property of least-cost design under constraints, not only a normative principle.

The optimisation also shows where the heuristic methods are sufficient and where they are not. The supply-only net-zero design of the linear programme coincides with that of the hourly design search, which the toolkit implements and which handles the monthly credit bank and the three billing rules explicitly. For shallow targets, sizing PV after efficiency and following the merit order therefore capture the main trade-offs. For deep targets, the joint optimisation is needed. It coordinates measures, PV, storage and curtailment hour by hour, and it reveals that the cost of the last tonnes of CO_2 is dominated by storage, which efficiency reduces but cannot remove. For SMEs facing emission targets from customers or lenders, this suggests a two-stage planning process. Simple tools can screen measures and size PV; an energy-system model such as urbs can then be used once storage or emission caps enter the decision.

5.6. Implications for practice and policy

For the plant, the results suggest a staged plan: (i) implement base-load control, PF correction and compressed-air maintenance immediately, as they are low-cost and verifiable; (ii) retrofit sewing motors and insulate heated equipment; (iii) size PV only after the savings have been measured; (iv) investigate thermal fuel use and heat recovery. The M&V plan prepared during the audit already specifies IPMVP options for each measure [57] and is summarised in Appendix E. For policy, three implications follow. First, distributed-generation programmes and green-credit lines aimed at SMEs could require an efficiency screening, such as a simple baseline regression and an audit, before PV sizing, in the spirit of the EU's EE1st guidance [3]. Second, tariff reforms that move GDMTO demand charges towards measured peaks would strengthen incentives for demand management but erode the value of PV. They should therefore be signalled in advance. Third, the large share of production-independent load points to the value of energy-management networks and capacity building for SMEs [14,25,26]. University research projects in which students apply energy-systems modelling to real decarbonisation problems are one route to building that capacity [83]; the open framework and toolkit presented here are designed to be used in such projects.

5.7. Transferability to other SMEs

The analysis rests on data that most SMEs already hold or can obtain in a walk-through audit (Table 8). Twelve months of bills and an end-use estimate are enough to rank measures against PV, to compare the supply-only and efficiency-first pathways and to compute the security indicators. A production series adds the baseline regression that identified the base load, the cheapest resource in this case. Hourly weather adds the net-zero analysis, and interval metering would replace the synthetic load profile, the weakest element of that analysis. The toolkit lists each input with its data status and its rank correlation with the NPV advantage of the efficiency-first portfolio across the Monte Carlo draws, so that a firm can see where the next measurement is worth making. Three results are likely to transfer: base-load control is cheap wherever production-independent load is large; efficiency first shrinks the PV array whenever the PV surplus is not remunerated; and storage is hard to justify under net metering. Others depend on the tariff. In particular, the regulatory risk to PV found here arises from Q-based demand charges; under a measured-peak tariff the toolkit assumes, conservatively, that PV earns no demand credit, whereas efficiency lowers the peak in proportion to its effect on average power, so efficiency would gain relative value.

Table 8. Data tiers for applying the method to other SMEs.

Data	Analyses enabled
12 monthly bills, tariff, end-use estimate, monthly climate	Tariff check, measure CCE and NPV, merit order, portfolios P1–P4, pathways, CO_2 and security indicators, Monte Carlo
+ monthly production	ISO 50006 baseline and base-load share
+ hourly weather year	Hourly PV, battery dispatch, net-zero and self-sufficiency frontier under three compensation regimes
+ interval metering	Measured always-on load in place of the assumed value (load shapes and time-of-use tariffs are not yet implemented)
+ fuel records	Scaling the per-GJ heat-electrification results to the plant

5.8. Limitations

This is an ex-ante evaluation study, and its limits should be read alongside its findings.

- *Data transcription.* Billing data were transcribed from utility bills rather than obtained from interval metering.
- *End-use split.* The split is an audit estimate for one month and is assumed constant across months.
- *Measure savings and costs.* Savings of E1–E4 and the costs of E2–E4 come from literature and market ranges rather than plant-specific measurements and quotations. The

Monte Carlo analysis shows that the conclusions hold across these ranges, but the absolute savings must be verified through the M&V plan.

- *Base-load evidence.* The baseline regression explains only part of the monthly variability, and the saving of base-load control, the main driver of the efficiency-first advantage, is taken from studies of other discrete-manufacturing processes. Behavioural and control savings may also decay without sustained energy management. Sub-metering and the M&V plan should therefore confirm this saving before the PV array is sized.
- *PV and load models.* The main PV model is monthly. The net-zero analysis uses hourly PV for one weather year rescaled to climatology and a synthetic hourly load built from an assumed always-on load and scheduled operating hours; shading and actual load shapes are not resolved, and inverter clipping is represented only in the EnergyPlus runs. Both models convert the NASA POWER hourly time stamps (UTC-7 at this longitude) to local clock time; the EnergyPlus weather file assumes constant relative humidity (60%) and wind speed (2 m s^{-1}), which affect module temperature only slightly. Storage sizing in particular should be confirmed with interval data.
- *Demand ceiling.* Measured maximum demand is only implied by the load factors printed on the bills. The model assumes that the energy-based demand term remains binding, which is consistent with the billed demand.
- *Tariff migration.* Migration to GDMTH, which would apply time-of-use energy prices and capacity charges on peak-period demand, is discussed but not simulated because no interval data were available; the bills identify the plant as a GDMTO customer, which is the tariff modelled.
- *Thermal fuel use.* Because fuel consumption was not metered, the fuel-substitution analysis is parametric.
- *Energy-security indicators.* Imported gas embodied in grid electricity is estimated with average shares and an assumed heat rate rather than with marginal dispatch.
- *Linear programme.* The urbs model optimises one representative year with perfect foresight, values grid electricity at a constant marginal price, represents net metering as annual netting without the 12-month expiry of credits, omits PV degradation and battery ageing and allows partial adoption of measures. It is used to test the design logic and to trace the decarbonisation frontier rather than to size equipment exactly.

None of these simplifications favours efficiency systematically. Several are conservative, notably the exclusion of non-energy benefits and the zero value assigned to PV surplus.

6. Conclusions

This study applied the energy efficiency first principle to the investment choices of a Mexican footwear SME, comparing demand-side measures with rooftop PV and heat electrification on common indicators of energy, emissions, cost and security.

The main conclusions are:

- *Validated tariff model.* A model built from official GDMTO charges reproduces the plant's bills with a mean absolute error of 2.2%. It shows that demand charges are effectively volumetric, so each avoided kWh is worth 2.64 MXN.
- *Cost of demand-side measures.* Base-load control, insulation of heated equipment, servo motors and a compressed-air package together save 23% of grid electricity. Their central costs of 0.21–1.04 MXN kWh^{-1} are below even the energy charge alone, and their paybacks are at most 2.2 years.
- *Sequencing and PV sizing.* PV is profitable, but sequencing matters. Implementing efficiency first achieves the same 54% reduction in grid purchases with 42% less PV capacity, 13% less capital and a higher NPV in 97% of uncertainty draws when the array is re-sized in each draw.
- *Security and resilience.* Efficiency lowers the plant's energy-implied billed demand, preserves its value if distributed-generation compensation changes and reduces exposure to imported gas and to energy-price shocks.

- *Heat electrification.* Heat pumps outcompete LPG boilers above a COP of 1.7, but not yet pipeline gas at current wholesale prices.
- *Net zero.* Annual net-zero electricity is achievable and profitable with 234–244 kWp of PV after efficiency (Python and EnergyPlus sizing; 69–72 kWp less than without it). Hourly self-sufficiency of 80% needs 350 kWp with 300 kWh of storage (nameplate) after efficiency and 90% turns the NPV negative; under current rules storage is a resilience investment, not a cost saving.
- *Joint optimisation.* In an urbs linear programme that co-optimises the measures, PV, storage and hourly operation, the efficiency bundle is completed before storage is added. Under emission caps of 35–10% of the baseline it lowers the annualised cost by 29–35% and delays and shrinks the battery, which dominates the cost of deep decarbonisation.

For SMEs in emerging economies, EE1st is best understood as a sequencing rule: measure, cut the waste, then size the supply. The open `ee1st` toolkit released with this paper allows other plants to repeat the assessment from twelve monthly bills and an end-use estimate, adding detail as production, weather or interval data become available. Future work should verify the savings ex post, add interval data to resolve hourly PV self-consumption and peak demand, and quantify thermal fuel use so that the phase-out of fossil fuels on site can be assessed directly.

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Abbreviations

The following abbreviations are used in this manuscript:

AC	Alternating current
CCE	Cost of conserved energy
CFE	Comisión Federal de Electricidad (Federal Electricity Commission)
COP	Coefficient of performance
CRF	Capital recovery factor
CV-RMSE	Coefficient of variation of the root-mean-square error
DC	Direct current
EE1st	Energy efficiency first
EnB	Energy baseline
EPW	EnergyPlus weather file
EU	European Union
GDMTH	Hourly large demand, medium-voltage tariff
GDMTO	Ordinary large demand, medium-voltage tariff
IPMVP	International Performance Measurement and Verification Protocol
LCOE	Levelised cost of electricity
LP	Linear programme
LPG	Liquefied petroleum gas
M&V	Measurement and verification
MXN	Mexican peso (kMXN, thousands; MMXN, millions)
NPV	Net present value
O&M	Operation and maintenance
PF	Power factor
PV	Photovoltaics
SME	Small and medium-sized enterprise
USD	United States dollar
UTC	Coordinated Universal Time
VAT	Value-added tax

Appendix A. Billing, production and tariff data

The monthly records were transcribed from utility bills and plant production records; they are confidential and only their summary is reproduced (Table A1). Table A2 lists the official tariff charges for Pachuca de Soto.

Table A1. Billing and production records, January 2024–April 2026 (summary only; monthly data confidential).

Variable	Months	Minimum	Mean	Maximum
Electricity (kWh per month)	28	19,040	35,523	64,160
Billed demand (kW)	28	46	89	157
Power factor (%)	28	84.4	87.1	90.6
Average price (MXN kWh^{-1}, pre-VAT)	28	2.770	3.005	3.153
Production (pairs per month)	25	3,473	13,692	29,460

Table A2. Official GDMTO charges, Centro Oriente division, pre-VAT [68].

Month	Fixed (MXN month^{-1})	Energy (MXN kWh^{-1})	Distribution (MXN kW^{-1})	Capacity (MXN kW^{-1})
2024-01	360.01	1.544	155.26	342.36
2024-02	360.01	1.552	155.26	344.34
2024-03	360.01	1.569	155.26	348.69
2024-04	360.01	1.569	155.26	348.70
2024-05	360.01	1.569	155.26	348.70
2024-06	360.01	1.569	155.26	348.70
2024-07	360.01	1.608	155.26	358.56
2024-08	360.01	1.621	155.26	361.67
2024-09	360.01	1.621	155.26	361.69
2024-10	360.01	1.621	155.26	361.70
2024-11	360.01	1.621	155.26	361.69
2024-12	360.01	1.597	155.26	355.67
2025-01	308.10	1.617	155.81	359.79
2025-02	308.10	1.620	155.81	360.36
2025-03	308.10	1.620	155.81	360.51
2025-04	308.10	1.620	155.81	360.51
2025-05	308.10	1.620	155.81	360.51
2025-06	308.10	1.654	155.81	369.07
2025-07	308.10	1.676	155.81	374.61
2025-08	308.10	1.678	155.81	375.13
2025-09	308.10	1.668	155.81	371.70
2025-10	308.10	1.668	155.81	371.70
2025-11	308.10	1.624	155.81	360.54
2025-12	308.10	1.553	155.81	342.52
2026-01	335.36	1.519	149.44	334.81
2026-02	335.36	1.520	149.44	335.00
2026-03	335.36	1.521	149.44	335.20
2026-04	335.36	1.521	149.44	335.19
2026-05	335.36	1.508	149.44	331.84
2026-06	335.36	1.495	149.44	328.54
2026-07	335.36	1.469	149.44	321.97
2026-08	335.36	1.469	149.44	321.98
2026-09	335.36	1.469	149.44	321.98
2026-10	335.36	1.469	149.44	321.98

Appendix B. Model parameters

Table A3 lists the model parameters. Ranges define triangular distributions whose mode is the central value. Status codes: M measured, O official, O* derived from official data, L literature, Q quotation in the audit report, D design, A assumption, R regulatory scenario. The last nine parameters belong to the hourly net-zero module (Section 3.9) and are varied in scenario analysis only.

Table A3. Model parameters, ranges and status.

Parameter	Central	Range	Status	Source / note
Real discount rate	0.1	0.06–0.14	A	Audit report; varied 6–14%
Appraisal horizon (yr)	20	fixed	A	Audit report
Real tariff escalation (yr^{-1})	0	-0.01–0.02	A	Constant real tariffs
Grid emission factor (tCO_2e MWh^{-1})	0.444	fixed	O	[21]
Gas-fired share of generation	0.638	0.6–0.68	O	[17] (Table 2.2)
Heat rate of gas-fired generation (GJ MWh^{-1})	7.2	6.5–8.5	A	Typical CCGT, 42–55% LHV efficiency
Imported share of natural gas	0.74	0.7–0.78	O*	[18] (6.4 of 8.6 Bcf d^{-1})
E1 saving (share of compressor energy)	0.2	0.1–0.3	L	[32]: leaks 20–30% reduced to <10%; 1% per 2 psi
E1 capital cost (MXN)	50000	35000–80000	Q	Audit report
E1 O&M (MXN yr^{-1})	8000	5000–12000	A	Annual ultrasonic leak survey
E1 life (yr)	10	fixed	A	
E2 clutch-motor share of stitching energy	0.7	0.5–0.85	A	
E2 saving per motor	0.65	0.5–0.75	L	[35]
Clutch motor running power (kW)	0.35	0.25–0.45	L	Literature range
E2 installed cost per servo motor (MXN)	3800	2800–5500	A	Market prices
E2 life (yr)	10	fixed	A	

Parameter	Central	Range	Status	Source / note
E3 heater share of vulcanising and injection energy	0.65	0.5–0.8	A	
E3 saving (share of heater energy)	0.2	0.1–0.3	L	[43]; grey literature 15–30%
E3 capital cost (MXN)	100000	60000–150000	A	Insulating jackets and controls
E3 life (yr)	8	fixed	A	
E4 saving (share of base load)	0.1	0.05–0.2	L	[36]; [37]
E4 capital cost (MXN)	40000	20000–70000	A	Timers, contactors, sub-meters
E4 life (yr)	10	fixed	A	
E5 capacitor bank (kVAr)	50	fixed	Q	Audit design
E5 power factor after correction	0.97	0.95–0.99	A	Automatic bank
E5 capital cost (MXN)	85000	70000–110000	Q	Audit report
E5 life (yr)	15	fixed	A	
S1 PV capacity (kWp)	163.8	fixed	D	252 modules of 650 W (audit design)
PV installed cost (MXN W^{-1})	11	9–14	Q	Audit quotation: 1.8 MMXN for 163.8 kWp
PV performance ratio	0.8	0.75–0.85	L	[76]
PV tilt (°)	20	fixed	D	Latitude tilt
Ground albedo	0.2	fixed	A	
PV degradation (yr^{-1})	0.005	0.003–0.008	L	Typical crystalline silicon
PV O&M (share of capital yr^{-1})	0.01	0.005–0.015	A	
Inverter replacement (share of capital, year 12)	0.1	0.08–0.15	A	
PV life (yr)	25	fixed	L	
Roof area (m^2 kWp^{-1})	5.5	5–6.5	A	650 W module of 3.1 m^2 with spacing
PV credit on billed demand (1 = yes)	1	0–1	R	Net metering reduces Q-based billed demand
LPG price (MXN kg^{-1}, incl. VAT)	19.76	fixed	O	CNE maximum price, PROFECO, 11–17 Oct 2026; VAT removed in the model
LPG lower heating value (MJ kg^{-1})	47.3	fixed	L	[77] (Vol. 2, Table 1.2)
LPG emission factor ($kgCO_2$ GJ^{-1})	63.1	fixed	L	[77] (Vol. 2, Table 2.2)
Natural-gas emission factor ($kgCO_2$ GJ^{-1})	56.1	fixed	L	[77] (Vol. 2, Table 2.2)
Natural-gas end-user markup over IPGN	2	1.5–3	A	Transport and distribution
Boiler efficiency	0.85	0.8–0.9	A	
Heat-pump COP	3	2.5–4	L	[51]; [52] ($<100\,°C$)
Recoverable compressor heat (share of input)	0.7	0.5–0.8	L	[32]: >80% available
Always-on load (kW)	15	8–25	A	Needs interval data
Battery installed cost (MXN per nameplate kWh)	8500	fixed	A	[79] pack 199 USD kWh^{-1} plus inverter and balance of system; break-even analysis
Battery round-trip efficiency	0.85	fixed	L	[79]
Battery usable fraction of nameplate	0.9	fixed	A	
Battery power (kW per nameplate kWh)	0.5	fixed	A	
Battery O&M (share of capital yr^{-1})	0.025	fixed	L	[79]
Battery life (yr)	15	fixed	L	[79]
Value of expired credits (MXN kWh^{-1})	0	0–1	A	0 = conservative
Export price under net billing (MXN kWh^{-1})	1	0.6–1.5	A	Stress case

Appendix C. Photovoltaic resource and yield

Monthly mean daily plane-of-array irradiation $\bar{H}_T$ was computed from NASA POWER monthly mean global ($\bar{H}$) and diffuse ($\bar{H}_d$) horizontal irradiation [73] as

$$\bar{H}_T = \bar{H}\left[\left(1-\frac{\bar{H}_d}{\bar{H}}\right)\bar{R}_b + \frac{\bar{H}_d}{\bar{H}}\,\frac{1+\cos\beta}{2} + \rho_g\,\frac{1-\cos\beta}{2}\right], \tag{A1}$$

with the monthly beam tilt factor $\bar{R}_b$ of Klein [75] for a south-facing surface evaluated on each month's recommended day, tilt $\beta = 20^\circ$ and ground reflectance $\rho_g = 0.2$. Annual mean horizontal irradiation is 5.05 kWh m^{-2} d^{-1}, compared with the 4.8 peak-sun hours used in the audit report, and plane-of-array irradiation is 5.23 kWh m^{-2} d^{-1}. With a performance ratio of 0.80 the specific yield is 1,526 kWh kWp^{-1} yr^{-1}. The audit report sized the array at 157.8 kWp to cover 80% of a 28,400 kWh month with 4.8 peak-sun hours and no performance ratio, and proposed 252 modules (163.8 kWp). This study adopts the module-based capacity.

Appendix D. Sensitivity analysis

Table A4 gives the 20-year NPV of each portfolio with each parameter at its low and high bound and all others at their central values, ranked by the swing of P4.

Table A4. One-way sensitivity of 20-year NPV (MMXN).

	P1 Efficiency		P2 PV only		P4 Efficiency first	
Parameter	low	high	low	high	low	high
Real discount rate	2.64	1.33	5.42	2.39	5.64	2.64
Real tariff escalation (yr^{-1})	1.68	2.22	3.23	4.34	3.45	4.62
E4 saving (share of base load)	1.56	2.40	3.57	3.57	3.53	4.36
PV installed cost (MXN W^{-1})	1.84	1.84	3.92	3.03	4.01	3.50
E3 saving (share of heater energy)	1.60	2.08	3.57	3.57	3.57	4.04
PV performance ratio	1.84	1.84	3.22	3.91	3.61	4.00
E2 saving per motor	1.64	1.97	3.57	3.57	3.61	3.94
E1 saving (share of compressor energy)	1.68	2.00	3.57	3.57	3.65	3.96
E2 clutch-motor share of stitching energy	1.69	1.95	3.57	3.57	3.66	3.92
E2 installed cost per servo motor (MXN)	1.93	1.69	3.57	3.57	3.89	3.66
E3 heater share of vulcanising and injection energy	1.73	1.95	3.57	3.57	3.70	3.91
Clutch motor running power (kW)	1.71	1.91	3.57	3.57	3.67	3.88

Appendix E. Measurement and verification plan

The audit team prepared an M&V plan following IPMVP [57]. PV output is verified under Option B through inverter and bidirectional-meter data with local irradiance. The capacitor bank is verified under Option C through billed PF and surcharges, or Option B where a network analyser is installed. Compressed air is verified under Option B through dedicated metering of kWh, pressure and operating hours, with specific consumption (kWh per m^3 of air) as the indicator where flow can be measured. Servo motors and insulation can be verified under Option A with spot power measurements and logged hours. Base-load control is verified under Option C through the production-normalised baseline of Section 3.3, complemented by night and weekend load readings. The whole facility is verified under Option C with at least 12 months of post-implementation billing, adjusted for production and operating hours. Given the CV-RMSE of the baseline (16.0%), savings below about 10% of consumption cannot be detected reliably at whole-facility level. Sub-metering of the compressed-air room and of the vulcanising and stitching areas is therefore recommended as the first implementation step.

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